

Reform of the Wildlife Act 1953

AN OPPORTUNITY FOR TRANSFORMATIONAL CHANGE OF AOTEAROA NEW ZEALAND'S BIODIVERSITY LAW

Appendix A

Comparative review of international
threatened species law



Environmental
Defence
Society

Photo by Bernard Spragg

Appendix A

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Abbreviations

ATK	Aboriginal Traditional Knowledge
Aotearoa	Aotearoa New Zealand
CCAA	Candidate Conservation Agreement with Assurances
COSEWIC	Committee on the Status of Endangered Wildlife in Canada
EPBC Act	Environment Protection and Biodiversity Conservation Act 1999 (Australia)
ESA	Endangered Species Act 1973 (U.S.)
IAC	Indigenous Advisory Committee
IPA	Indigenous Protection Areas
IPBES	Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services
IUCN	International Union for Conservation of Nature
NACOSAR	National Aboriginal Council on Species at Risk
NPSIB	National Policy Statement on Indigenous Biodiversity
NWT	Northwest Territories
NZTCS	New Zealand Threat Classification System
Ontario’s ESA	Endangered Species Act 2007 (Ontario)
SARA	Species at Risk Act 2002 (Canada)
TECs	Threatened Ecological Communities
TSSC	Threatened Species Scientific Committee
U.S.	United States of America

Introduction

Appendix A is a comparative review of international threatened species law, focusing on three key jurisdictions: Australia, Canada and the United States (U.S.) Over 30 international experts from the U.S., Canada, Australia, Scotland and Sweden were interviewed for Appendix A, and provide insights into how international threatened species legislation is performing.

A note on the term ‘threatened’

This Appendix refers to threatened species and threatened species laws. The term Threatened encompasses the three highest threat categories of the New Zealand Threat Classification System (NZTCS) (Nationally Critical, Nationally Endangered and Nationally Vulnerable). Generally, however, unless it is clear that the Appendix is referring to species listed as Threatened in Aotearoa New Zealand (Aotearoa), when it refers to threatened species Appendix A is referring more broadly to species that can be listed in any threat category under overseas laws or internationally accepted methodologies (e.g., the International Union for Conservation of Nature (IUCN) Red List). 'Threatened species law' is a general descriptor that this Appendix uses to refer to legal frameworks for the listing, protection and recovery of species, across all threat categories.

Listing categories and criteria

Under international threatened species laws, the level of protection afforded to a species, and the extent to which recovery actions are required for that species, depend on its threat classification. The conservation status of a species is assessed (by an extinction risk assessment), and that species is 'listed' in a corresponding category reflecting its risk of extinction (its threat classification or category).

Species listed in higher threat categories, such as Endangered or Threatened, receive more protection and greater recovery support than less imperiled species. While the higher threat categories often attract more attention because of their regulatory implications, lower threat categories should not be overlooked as they can play an important role in slowing species decline.

Australia, Canada and the U.S. all have different categories of listed species that are protected by legislation. Differences lie largely in the criteria used for each category, and how far along the risk continuum nations go to protect species that are not yet Endangered or Threatened but are of future concern.

The U.S. approach

Under the Endangered Species Act 1973 (ESA), species can only be listed as Endangered or Threatened. Threatened has a high threshold of likely to become Endangered in the foreseeable future. Discretionary terms such as likely and foreseeable have caused issues when applying the Act and listing species as Threatened.

The U.S. does not use the IUCN's Red List for listing species, rather it has its own set of five criteria that are set out in the ESA. Under that Act, a species can be classified by regulations as Endangered or Threatened because of:¹

- The present or threatened destruction, modification, or curtailment of its habitat or range;
- Over utilisation for commercial, recreational, scientific, or educational purposes;
- Disease or predation;
- The inadequacy of existing regulatory mechanisms; or
- Other natural or manmade factors affecting its continued existence.

These assessment criteria focus on threats and effectiveness of existing protections. They do not include intrinsic factors that increase extinction risk (e.g., small population size or small habitat range) which are factored into Canada's and Australia's species assessments.

Use of IUCN categories and criteria in Canada and Australia

¹ Endangered Species Act 1973, s 4

Canada and Australia use the IUCN Red List categories and criteria for listing species. The threat categories adopted by each jurisdiction are set out in primary legislation, but the criteria for assessment are set out in policy.

To inform this Appendix, the Chairs of the Committee on the Status of Endangered Wildlife in Canada (COSEWIC) and Australia's Threatened Species Scientific Committee (TSSC) (which undertake species extinction risk assessments) were interviewed. Both supported use of the IUCN's criteria, which are seen as the international gold standard for assessing the extinction risk of a species. The IUCN has undertaken a significant amount of work developing, interpreting, and standardising the assessment criteria, allowing for international comparisons. The IUCN also has several guidance resources and runs training programmes on using the criteria.

The NZTCS uses a modified version of the IUCN's Red List system that reflects Aotearoa's small size and high endemism. If the IUCN criteria were applied without modification in Aotearoa, many more species would be classified as Threatened because the criteria poorly reflect the capacity for island species to persist as small, isolated populations. Aotearoa has many such species.

The IUCN Red List system has nine extinction risk categories: Extinct, Extinct in the Wild, Critically Endangered, Endangered, Vulnerable, Near Threatened, Least Concern, Data Deficient and Not Evaluated.

Canada has adopted Endangered and Threatened and has a Special Concern category. The latter is akin to the IUCN's Near Threatened status. By listing species of Special Concern, Canada goes further than the U.S. in terms of listing species that are at-risk, but not yet Endangered or Threatened. Under Canada's Species at Risk Act 2002 (SARA) a species of special concern is one that may become Threatened or Endangered because of a combination of biological characteristics and identified threats. The Minister must prepare a management plan for such species and its habitat which, must include measures for the conservation of the species. No recovery strategy or recovery action plan is required for species of special concern.

Other than the requirement for a management plan, the Special Concern category has no regulatory impact. The prohibited acts and habitat conservation provisions of SARA do not apply to this category.

Five IUCN quantitative criteria are used to assess a species' status in Canada:²

- Decline in total number of mature individuals
- Small distribution range and decline fluctuation
- Small and declining number of mature individuals
- Very small or restricted total Canadian population
- Population projections

Each criterion has thresholds to guide assessment. Figure 1 is an example for the first criterion: number of mature individuals. Criteria A1 was specifically inserted to address commercial fish species.

² <https://cosewic.ca/index.php/en-ca/assessment-process/wildlife-species-assessment-process-categories-guidelines/quantitative-criteria.html>

A. Decline in Total Number of Mature Individuals

Indicator	Endangered	Threatened
A1. An observed, estimated, inferred or suspected reduction in total number of mature individuals over the last 10 years or 3 generations, whichever is the longer, where the causes of the reduction are: clearly reversible and understood and ceased, based on (and specifying) any of the following ¹ :	Reduction of ≥ 70%	Reduction of ≥ 50%
(a) direct observation		
(b) an index of abundance appropriate to the taxon		
(c) a decline in index of area of occupancy, extent of occurrence and/or quality of habitat		
(d) actual or potential levels of exploitation		
(e) the effects of introduced taxa, hybridization, pathogens, pollutants, competitors or parasites.		
A2. An observed, estimated, inferred or suspected reduction in total number of mature individuals over the last 10 years or 3 generations, whichever is the longer, where the reduction or its causes may not have ceased or may not be understood or may not be reversible, based on (and specifying) any of (a) to (e) under A1.	Reduction of ≥ 50%	Reduction of ≥ 30%
A3. A reduction in total number of mature individuals, projected or suspected to be met within the next 10 years or 3 generations, whichever is the longer (up to a maximum of 100 years), based on (and specifying) any of (b) to (e) under A1.	Reduction of ≥ 50%	Reduction of ≥ 30%
A4. An observed, estimated, inferred, projected or suspected reduction in total number of mature individuals over any 10 year or 3 generation period, whichever is longer (up to a maximum of 100 years in the future), where the time period must include both the past and the future, and where the reduction or its causes may not have ceased or may not be understood or may not be reversible, based on (and specifying) any of (a) to (e) under A1.	Reduction of ≥ 50%	Reduction of ≥ 30%

Figure 1. Criteria for assessing threat status based on number of mature individuals³

Contextual considerations are then reviewed and, if thought to be significant, may modify the initial quantitative assessment. Such considerations include the rescue potential from outside of Canada and other life-history characteristics that may not have been adequately assayed (e.g., age and size at maturity, dispersal strategy and longevity).

In Australia, the Environment Protection and Biodiversity Conservation Act 1999 (EPBC Act) provides for the listing of Endangered, Critically Endangered, Vulnerable and Conservation Dependent species. The latter is not an IUCN category. Under the EPBC Act, native species can be listed as Conservation Dependent if cessation of a conservation programme would result in the species becoming Critically Endangered, Endangered or Vulnerable. In this regard, Australia also goes further than the U.S. in listing species not yet Endangered or Threatened.

The EPBC Act imposes no regulatory protection for species listed as Conservation Dependent (they are not currently matters of national environmental significance (see below *Spotlight: The EPBC Act's operation*), and enforcement action cannot be taken in relation to harm inflicted on species in this category. Some environmental groups in Australia dislike the conservation dependent category, considering it 'cutting of green tape' - a way of avoiding the listing of a species.

Spotlight: The EPBC Act's operation

The EPBC Act is a wide-ranging Act providing species and habitat protection but also covering planning matters. Under the Act, matters of national environmental significance are protected and include species that are extinct in the wild, critically endangered species, endangered species, vulnerable species, critically endangered communities and endangered communities.

An activity that has, or will have, a significant impact on a matter of national environmental significance triggers the Act and must undergo a full assessment. Significant impact on species listed as Conservation Dependent and key threatening processes (discussed below) does not trigger the Act.

³See table A at <https://cosewic.ca/index.php/en-ca/assessment-process/wildlife-species-assessment-process-categories-guidelines/quantitative-criteria.html>

While the category has broad application to native species at risk, in practice, only commercially fished marine species and their bycatch are listed as Conservation Dependent.

In some circumstances, a conservation dependent listing may produce better outcomes for a species than being listing in higher threat categories. Unlike other listings, conservation dependent species must already be subject to a conservation programme or a management plan at the time of listing (as opposed to a recovery plan being developed after listing). This requirement usually means that the species is already being monitored e.g., catch data (used to measure the success of fisheries management strategies) provides a continuous means of monitoring fish populations. For by-catch species, deaths are monitored because everything is landed. Listing a fish or bycatch species in a higher threat category may result in a reduction of monitoring and lead to a paucity of information as to the health of the population. The issue is illustrated in the case of the scalloped hammerhead shark (see below *Spotlight: The conservation dependent scalloped hammerhead shark*).

Spotlight: The conservation dependent scalloped hammerhead shark

The scalloped hammerhead shark is a bycatch of some Australian in-shore commercial fisheries. It is currently listed as Conservation Dependent but qualifies for listing as Endangered. The biggest issue for the shark is incidental by-catch mortality. Currently, shark by-catch is landed, which enables catch monitoring. If the species' threat status were raised, landing would no longer be allowed, those caught would be discarded at sea and not recorded.

The TSSC is currently reviewing the shark's listing status. While the outcome of that assessment is not yet known, the Committee's recent submission on the EBPC Act provides an insight into what its preferred recommendation might have been (if the law enabled it).

In its submission, the TSSC recommended that the conservation dependent category be deleted, that species instead be listed in the threat category to which they qualify, and that a plan of management and a monitoring and assessment plan be implemented. For the scalloped hammerhead shark, this would mean an endangered listing with a special management 'flag'. That flag would exempt the shark from being a matter of national environmental significance (thus it would not trigger the Act). But, the Committee submit, that 'flag' would be removed if the plan of management is no longer operational or fails to recover the species. If that occurred, the shark would become a matter of national environmental significance (by virtue of its endangered status) and trigger the Act. This approach could incentivise fishers to sustainably manage species. It might also usefully be applied to some terrestrial situations.

At the State level, Western Australia's Biodiversity Conservation Act 2016 includes a listing category for species of special conservation interest. This category applies to species that are not threatened but:

- Have a naturally low population;
- Have a restricted natural range;
- Are subject to or recovering from a significant population decline or reduction in natural range; or
- Are of special interest to science.

Western Australia also has a catch-all listing category: species otherwise in need of special protection. This captures species not caught by the above category, but which need protection. Currently, only seven species are listed as otherwise in need of special protection.

Like Canada, Australia uses the IUCN's five criteria and associated quantitative thresholds for its extinction risk assessments (Figure 2). While some of the terminology differs between Canada and Australia, the two assessments use the same thresholds for each of the criteria. All but two Australian States (Western Australia and South Australia) have agreed to align with this federal method of assessment.

Criteria for listing threatened species (Part 7.01 of the EPBC Regulations)			
Criterion	Critically Endangered	Endangered	Vulnerable
1. It has undergone, is suspected to have undergone or is likely to undergo in the immediate future:	a <u>very severe</u> reduction in numbers	a <u>severe</u> reduction in numbers	a <u>substantial</u> reduction in numbers
2. Its <u>geographic distribution is precarious</u> for the survival of the species and is:	<u>very restricted</u>	<u>restricted</u>	<u>limited</u>
3. The estimated total number of mature individuals is:	<u>very low</u>	<u>low</u>	<u>limited</u>
and either of (a) or (b) is true:			
(a) evidence suggests that the number will continue to decline at:	a <u>very high</u> rate	a <u>high</u> rate	a <u>substantial</u> rate
(b) the number is likely to continue to decline and its geographic distribution is	<u>precarious</u> for its survival	<u>precarious</u> for its survival	<u>precarious</u> for its survival
4. The estimated total number of mature individuals is:	<u>extremely low</u>	<u>very low</u>	<u>low</u>
5. The probability of its extinction in the wild is at least	50% in the <u>immediate</u> future	20% in the <u>near</u> future	10% in the <u>medium-term</u> future

Figure 2. Criteria for listing species in the Critically Endangered, Endangered or Vulnerable categories under the EPBC Regulations⁴

In practice, the criteria are applied as follows in Australia:

- Criterion 1: Population size reduction - species listed only under this criterion are likely to be reasonably common and widespread but threatened by something occurring across much of their range. It follows that piecemeal conservation efforts at a site level are unlikely to be effective, and a species-wide response will be required. Longline fishing threats to seabirds is a good example of this working in practice.
- Criterion 2: Geographic distribution as indicators for either extent of occurrence and/or area of occupancy - species falling only within this criterion are likely to have a small and declining range. Any habitat is likely to be critical habitat, and development should be assessed with rigor as any loss of habitat is likely to exacerbate extinction risk.
- Criterion 3: Population size and decline - threshold rates of decline are lower for these species than for the first criterion above (population size reduction) so that even small site-based conservation action will be important.
- Criterion 4: Number of mature individuals - species listed only under this criterion have very small total populations, so every individual is important for the species' survival. Higher thresholds and penalties could be imposed for adverse human interactions with these species.⁵

⁴ Environmental Protection and Biodiversity Conservation Regulations 2000, part 7

⁵ Garnett, 2013, 17-18

The IUCN's criteria have long been the most widely used tool for measuring extinction risk. However, the approach has limitations for identifying species at the very edge of extinction. For example, the IUCN categories are not designed to distinguish species on a rapid trajectory towards extinction from those with very small populations that may persist for long periods, even though their extinction risk is high should circumstances change. This situation arises in Aotearoa for fauna confined to islands.

Scientific experts in Australia have suggested that the status of a listed species, based as it is on the IUCN's criteria, may therefore be a poor predictor of future risks, especially if novel threats are encountered, or if existing threats rapidly become magnified due to unprecedented events (e.g., the 2019 - 2020 Australian wildfires). Experts suggest there may be value in undertaking additional assessment to more explicitly estimate extinction risk over time and therefore accurately ascribe threat status to species.

Factoring climate change into extinction risk assessments

The potential impact of climate change on biodiversity is well documented. While the IUCN's Red List guidance document discusses adopting a precautionary approach in the face of uncertainty, it does not specifically mention climate change.

Climate change *per se* is not well addressed in international extinction risk assessments. The IUCN has a climate change vulnerability assessment tool, but COSEWIC does not have the resources to apply it. In Australia, the national science programme has a project underway that will better enable climate risk information to be factored into extinction risk assessments. Generally, the issue in Australia with factoring in climate change is obtaining information and agreed timeframes at the small spatial scale. Under the NZTCS, climate change is considered in the listing process through the climate change qualifier.⁶

Factoring climate change into extinction risk assessments and other aspects of threatened species law is important, as is building biodiversity resilience to climate change.

The Bramble Cay melomys (*Melomys rubicola*) is the first mammal reported to have gone extinct as a direct result of climate change. Previously found only on the island of Bramble Cay in Great Barrier Reef, its habitat was destroyed by rising sea levels.⁷

Data deficient species

Australia does not have a Data Deficient threat category (which the IUCN does). Instead, when a species being assessed by the TSSC has insufficient data to be assigned a threat category it is found ineligible for recommendation (although for transparency these species are listed on the TSSC's website as data deficient).

The IUCN defines the Data Deficient category as:

"A taxon is Data Deficient when there is inadequate information to make a direct, or indirect, assessment of its risk of extinction based on its distribution and/or population status."

⁶ NIWA has undertaken a climate change vulnerability assessment for fish, see <https://niwa.co.nz/te-kuwaha/ccva>. DOC has started similar assessments for terrestrial species (birds, bats, invertebrates and vascular plants). These assessments will be finalised in the next year or so.

⁷ <https://www.iucn.org/resources/issues-briefs/species-and-climate-change>

It states that “a taxon in this category may be well studied, and its biology well known, but appropriate data on abundance and/or distribution are lacking”.

The TSSC guidelines state that data deficient species could include wide ranging species where information is only available on impacts and populations across part of their range. It notes that in some cases, the available information cannot easily be extrapolated across the entire range and therefore it is hard to determine i.e., whether a decline in one part of the range represents trends across the whole range. Without information across the national extent of the species, justification against the extinction risk criteria is difficult.

Because the EPBC Act does not recognise data deficient species there is no regulatory protection afforded to such species, nor any monitoring requirement mandated. Several plant species in Australia have been assessed as ineligible for recommendation. After the wildfires, the TSSC became increasingly concerned about these species. It is currently working with federal agencies to change policy so that if a species is data deficient, conservation advice can still be formulated for it. While conservation advice holds little regulatory weight, the hope is that they will at least help fill the information gap for such species.

In Western Australia, species that may possibly be threatened, but do not meet the criteria for listing because of insufficient scientific knowledge, are assigned a ranking and placed on a priority list. Species on that list are prioritised for monitoring and subsequently evaluation for listing.⁸

As noted by the IUCN, great care should be exercised in choosing between the Data Deficient category and a threatened status. Considerable uncertainty often exists with species data, often resulting from natural variability, semantic uncertainty and measurement error. The way in which this uncertainty is handled can strongly influence a species’ extinction risk assessment. The IUCN recommends methods for dealing with uncertainty. It states that assessors should resist an “evidentiary attitude and adopt a precautionary but realistic attitude” to uncertainty when applying the criteria. Overall, the IUCN states that a single listing category must be chosen, and it should be precautionary and credible.

The liberal use of the Data Deficient category is commonly pursued by vested interests trying to prevent the threat status of a species being properly recognised. But a lack of information on a species does not automatically relegate it to a Data Deficient category. The IUCN’s guidance on the use of the Data Deficient category states:

“It is important to recognize that taxa that are poorly known can often be assigned a threat category on the basis of background information concerning the deterioration of their habitat and/or other causal factors; therefore the liberal use of ‘Data Deficient’ is discouraged.”

In Canada, species can be listed as Data Deficient (although there is a preference for not assessing species without requisite science). When the issue does arise, COSEWIC is often left having to decide between Data Deficient or Endangered categories.

Incorporating indigenous knowledge in extinction risk assessments

Australia and Canada both have processes to incorporate indigenous knowledge into extinction risk assessments.

⁸ Department of Biodiversity and Attractions, 2019

The preamble of SARA states that the traditional knowledge of the aboriginal peoples of Canada should be considered in the assessment of which species may be at risk. In that regard, COSEWIC is required to carry out its functions based on the best available aboriginal traditional knowledge.

COSEWIC has an aboriginal traditional knowledge (ATK) subcommittee, which is mandated by SARA. The Chairperson and members of the subcommittee are appointed by the Minister after consultation with any aboriginal organisation the Minister considers appropriate. The subcommittee is meant to function as a conduit between Canada's aboriginal peoples and the extinction risk assessment process, bringing aboriginal knowledge to the table.

At the time of its creation, the ATK subcommittee was seen as world leading. However, while it has had some successes using aboriginal knowledge to influence listing decisions (e.g., the polar bear and the chinook salmon), these examples are the exception. Generally, the subcommittee has not performed as it was intended to. There are multiple reasons for that.

- It is difficult to source aboriginal knowledge. There are challenges around who owns that knowledge, who can share it and with whom, and how that knowledge is protected.
- Protections afforded to a listed species can have significant implications for aboriginal peoples (e.g., if a community cannot hunt that species anymore, or use that species' pelt as a source of clothing). Historically, the implications of listing were not well communicated to indigenous peoples which led to confusion about what could and could not be done with respect to listed species. Consequently, some aboriginal peoples are now unwilling to engage in extinction risk assessment processes.
- The ATK subcommittee is expected to be fully involved in each assessment process but has struggled to keep up with the volume of species being assessed, and sourcing aboriginal knowledge for each one. It is expensive to arrange and undertake conversations with aboriginal peoples, asking them to share their knowledge, and the subcommittee does not have the resources or capacity to do that. This has necessitated the ATK subcommittee prioritising species that aboriginal peoples might be interested in.
- There are regional differences among aboriginal peoples which means that national level threat assessments are inappropriate.
- It is difficult to transfer aboriginal knowledge into a scientific threat assessment, and have that knowledge given equal weight. Chairs of COSEWIC and the ATK subcommittee described aboriginal knowledge as a 'different way of knowing' and noted that it should not be contextualised into scientific models.

What is indigenous knowledge?

The ATK subcommittee has described indigenous knowledge as:

- Indigenous Knowledge Sciences are relationship based to include information on ecological relationships between humans and their environment: characteristics of species, habitat, and locations.
- Laws and protocols for human relationships with the environment are passed on through teachings and stories; in ceremony; in the Indigenous languages; and can be based on long-term observations.

- Place names provide information about harvesting areas, ecological processes, spiritual significance or the products of harvest.
- Indigenous Knowledge Sciences can identify life history characteristics of a species or distinct differences between similar species.

Based on COSEWIC's criteria, indigenous knowledge is:

"... species occurrence, distribution, population size and observed changes in biology, distribution and/or population size, over time; potential threats; existing management plans; harvesting restrictions, Aboriginal name and dialect and other relevant information."

Specific information regarding aboriginal cultural, medicinal or spiritual use of a species is not included in species status reports. Intangible cultural values do not form part of Australia's threat assessment either. Globally, indigenous peoples are seeking the recognition of plants and animals which are of cultural significance known as Cultural Keystone Species. In Australia, this is manifesting in recognition of culturally significant entities, which encompass both species and ecosystems. Provision for these entities is separate, and in addition, to the protection of threatened species.

As a result of these challenges, the ATK subcommittee is developing a new approach to its processes and protocols. Initial indications are that the subcommittee is likely to shift its focus to an ecosystem scale watershed approach, examining major threats to species of particular concern to aboriginal peoples, prioritising species of particular interest to them, and engaging with regional elder circles.

To inform a new direction, the subcommittee is undertaking workshops across Canada with aboriginal peoples and elder circles to get their recommendations. The subcommittee plans to produce a report on the matter. EDS highly recommends that the outcomes of that report be followed up.

Australia does not have an analogous indigenous statutory body informing its extinction risk assessments. Consequently, indigenous knowledge is poorly fed into threat classification decisions on an *ad hoc* basis.

There is emerging discussion in Canada and Australia about whether there needs to be two separate extinction risk assessment reports for each species to ensure that indigenous knowledge is appropriately presented and considered. One report based on scientific knowledge, and one based on community and indigenous knowledge.

In early 2021, the Northwest Territories (NWT) in Canada changed its processes to do just that. Now each extinction risk report has two parts, an indigenous and community knowledge component and a scientific knowledge component.⁹ The intent is that a complete status report will include both components (unless the NWT Species at Risk Committee determine that there is insufficient information available to complete one component). The new approach is bicultural and moves away from the idea of integrating indigenous knowledge into extinction risk assessments, to acknowledging that indigenous knowledge and science are equal in their integrity as knowledge systems.

Interviewees with expertise in indigenous knowledge expressed support for a dual reporting approach. Often described as the 'two-eyed seeing' method, whereby science and indigenous knowledge are both respected.

⁹ See, Species at Risk Committee, 'Detailed instructions for preparation of a Sarc Status Report: Indigenous and community Knowledge component', https://www.nwt-speciesatrisk.ca/sites/enr-species-at-risk/files/sarc_instructions_ikck_final_may420.pdf

Whatever approach is adopted, it is important that both knowledge systems are drawn on in a manner that prevents consideration of economic matters and the influence of vested interests.

Indigenous advisory bodies

Canada and Australia both have indigenous advisory bodies established to inform the Minister on threatened species related matters.

Canada has three Indigenous Peoples: First Nations, Inuit, and Metis. SARA establishes a National Aboriginal Council on Species at Risk (NACOSAR) to advise the Minister on the administration of the Act and to provide advice and recommendations to the Canadian Endangered Species Conservation Council (made up of several Ministers).

The NACOSAR has been met with criticism for being ineffective and inaccessible.¹⁰ This is partly to do with under-resourcing. NACOSAR lacks funding, staff, and even its own premises. NACOSAR also lacks control over its own administrative budget, agenda, and the work program it can undertake. These are instead controlled by public servants, which frustrates NACOSAR's ability to directly advise the Minister.¹¹ Indigenous participation has also been hindered by a lack of guidelines on the engagement process.¹²

In Australia, the EPBC Act sets up an Indigenous Advisory Committee (IAC) which advises the Environment Minister. An independent review of the EPBC Act recently found that the IAC "exemplifies" the "culture of tokenism" inherent in the Act. This is due to the discretionary and optional nature of its advisory function. Committee involvement is informal and *ad hoc* rather than formal and clearly defined. Indigenous membership on the Committee is sometimes the extent of integration of indigenous knowledge. The IAC also lacks adequate funding.¹³

The TSSC recently recommended that the IAC's role be expanded to include the listing of culturally significant entities and the preparation of statutory advice on how each entity could be conserved.¹⁴

Missed opportunity of pre-listing conservation

If a species is not listed in a threat category, its protection is largely left to general wildlife and animal welfare laws. This is a weakness of the U.S. and Canadian systems because when a species is not listed, its protections, particularly at the federal level, are extremely weak. Any protection is reliant almost entirely on state and territorial hunting rules. Species are left to drift towards endangerment.¹⁵ There are many species in Australia, the U.S. and Canada that have not been listed under national or federal legislation and have continued to decline.¹⁶

Not only is this an extremely bad conservation outcome, but it is bad economics. Wildlife recovery actions are usually less expensive and more effective when implemented before a species reaches crisis levels.¹⁷ This is the missed opportunity of pre-listing conservation.¹⁸ There are objective tools that can assess the likelihood of a species becoming threatened. These could be employed to

¹⁰ Johnson, 2010, *Submission to the Standing Committee on Environment and Sustainable Development*, 22 April 2010

¹¹ McNeely et al, 2011, 30

¹² Turcotte et al, 2021, 1474-1494, 1484

¹³ Samuel, 2020, 32

¹⁴ Marsh, 2020, *Submission to the EPBC Act review*, Threatened Species Scientific Committee, March 2020

¹⁵ See Holmer, 2016

¹⁶ Kraus et al, 2021, 1088-1127, 1093

¹⁷ Kraus et al, 2021, 1088-1127, 1095

¹⁸ Kraus et al, 2021, 1088-1127, 1095

promote early conservation interventions.¹⁹ Early identification of species at-risk is particularly important in the context of climate change, as evidenced in the below *Spotlight: The koala story*.

Incentivising private landowners to protect and recover species is an important component of pre-listing conservation and is discussed separately below.

Spotlight: The koala story

In 2011, Australia's Environment Committee recommended that the Government establish a nationally coordinated and integrated program for population monitoring of threatened species and other culturally, evolutionary and/or economically significant species. The recommendation was made in the context of the EPBC Act's reactive listing process that is ill-suited to koala conservation needs. The Committee proposed monitoring species of cultural, evolutionary and/or economic significance and a new category of listed species called Nationally Significant.

In response to the Committee's recommendations the koala was listed as Vulnerable in 2012. In February 2022, that classification was changed to Endangered for Queenstown, New South Wales and the Australian Capital Territory. Bushfires caused a dramatic decline in koala populations in these areas (although land clearance has also been blamed). Increased frequency and intensity of wildfires are a known consequence of climate change. Early recognition of this risk could have facilitated more proactive protections for the koala, which has gone from not being listed, to being listed as Endangered in some States, in just one decade.

Lessons learnt: Listing categories and criteria

1. Wildlife conservation undertaken prior to a species becoming at risk or threatened is important for slowing or preventing species decline.
2. The IUCN Red List system a robust risk extinction tool and has been usefully modified to reflect Aotearoa's circumstances.
3. Lesser threat categories can be usefully applied to generate monitoring and management responses to prevent species from becoming more threatened.
4. Commercial fish and fisheries bycatch species may need a more nuanced listing approach to reflect the interface between fisheries legislation and threatened species law.
5. A data deficient classification could usefully trigger a prioritised monitoring response.
6. Current thinking indicates that it is best to have dual reporting in extinction risk assessments, for indigenous knowledge and scientific knowledge.
7. Protections are still required for species not listed under threatened species law.

The listing process

Prioritisation of what species to assess

In an ideal world, all species would be afforded an equal opportunity to be assessed for their extinction risk and potentially listed. Internationally, that is rarely the case in practice. Large charismatic species or those with a high public profile tend to dominate international threatened species lists, with fungi, invertebrates and non-vascular plants taking a back seat. Lack of funding and knowledge deficits have been blamed for the taxonomic bias. So too has the lack of strategic identification of species that require assessment. Uneven species coverage has led to an uneven allocation of conservation efforts.²⁰

¹⁹ Kraus et al, 2021

²⁰ Walsh et al, 2012, 134 -143

While Canada suffers from species bias, it does so to a lesser extent than the U.S. or Australia, primarily because of the structure of its scientific committee. At the federal level in Canada, COSEWIC is responsible for identifying species for an extinction risk assessment and has ten species specialist subcommittees each relating to a taxon: amphibians and reptiles, arthropods, birds, freshwater fishes, marine fishes, marine mammals, molluscs, mosses and lichens, terrestrial mammals and vascular plants. While this is a comprehensive taxa list, fungi, slime moulds, algae, worms, coelenterates and echinoderms all still miss out.

Annually, COSEWIC's subcommittees produce a list of candidate species for extinction risk assessment. Each species is assigned a priority ranking according to levels of apparent risk and with consideration to taxonomic distinctness, global distribution and proportion of range within Canada. Only biological factors are used to prioritise species (issues such as adequate scientific knowledge are not considered at this stage).

In Australia, nomination of individual species (usually by the public) has been *ad hoc*, with a significant bias toward charismatic vertebrates. Low profile species have suffered in this system, including the now extinct forest skink, which went extinct only four months after being identified as Critically Endangered under the EPBC Act.²¹

The U.S. system includes a category of candidate species that is 'warranted but precluded'. These are species that qualify for assessment as Endangered or Threatened but developing a proposal for listing is precluded by other, higher priority species. Species that are warranted but precluded have no statutory protection under the ESA. While waiting for a listing assessment, federal agencies can develop a Candidate Conservation Agreement with Assurances (CCAA) for the species (see below *Spotlight: Candidate Conservation Agreement with Assurances*).

In Aotearoa, the NZTCS assesses all documented indigenous species as much as possible (in cyclic rotations for each taxon). This process should continue to avoid bias issues discussed. We do find it interesting however that the NZTCS is able to assess so many species, while internationally, analogous independent bodies struggle to keep up with assessments and have prioritised species. For the reasons discussed next, it is important that Aotearoa's extinction risk assessment process is scientifically defensible.

Importance of a robust list

Species lists have two fundamental impacts; on the potential survival of a species and on potential land uses that might harm a species or its habitat.

A biased or poorly conceived list compromises its credibility and promotes cynicism about its contribution to effective conservation.²² This would undermine constructive conservation effort among private landowners and local authorities. If credibility in a threatened species list is lost, it may also become easier for land users to obtain exemptions to species regulations.

It is important therefore that species lists comprehensively cover all species, accurately reflect species' extinction risk and are devoid of political or vested interest interference. The whole process needs to be transparent, efficient and undertaken by independent scientific assessors.

²¹ Woinarski et al 2016, 13-23.

²² Garnett, 2013, 13

Internationally, once an extinction risk assessment has been undertaken, the final decision to ‘list’ a species in a threat category is political. The high level of Ministerial discretion provided in the EPBC Act and SARA is often criticised for making the listing process opaque and significantly delayed. The main problem being that this provides unfettered opportunity for vested interests to block the conservation actions needed to prevent further decline and loss.

In Aotearoa, species are automatically listed in a threat category based on their extinction risk assessment. Separate Ministerial approval to ‘list’ the species is not required. We heard some criticisms of the NZTCS process, with some interviewees calling for it to be regulated. Others worried that regulating the process would expose it to political and interest group influence. The following sections of this Appendix describe the statutory listing processes of the U.S., Canada and Australia. They are included as a caution for Aotearoa if it was to regulate its process. As detailed, overseas systems have experienced significant problems because of political and vested interest interference in the process of ascribing a threat status to a species. If extinction risk assessments and listing of species in threat categories under the NZTCS was to be regulated, the statutory framework would need to be carefully crafted to ensure that the process was insulated from external pressures. An important aspect of that is ensuring automatic listing of species into threat categories (discussed below).

Listing process in the U.S.

Listing decisions in the U.S. are heavily influenced by scientific knowledge which is collated in biological risk reports called Species Status Assessments.²³ Such assessments detail the species’ habitat and demographics and the probable explanations for past and ongoing changes in abundance and distribution, and forecast the species’ response to future environmental conditions and conservation efforts.

Information obtained on a candidate species is peer reviewed and public comment is sought before the US Fish and Wildlife Service, or the National Marine Fisheries Service for marine mammals, (hereinafter, the ‘federal agencies’) makes its listing decision.

The federal agencies are required to make listing decisions based solely on the best scientific evidence and commercial data available after taking into account efforts to protect the species. Economic considerations are meant to be irrelevant to the listing process (but can be considered when determining critical habitat designations).²⁴

Notwithstanding, the federal agencies (which serve the public interest and the President’s agenda) can reject scientific findings or peer reviews so long as the decision is not arbitrary and capricious (which is the standard for judicial review in the U.S.). Thus, although listing decisions in the U.S. are informed by extensive scientific evidence, politics run interference in the process.

Under the ESA, listing decisions are meant to take two years. On average it has taken the US Fish and Wildlife Service 12 years. Cumbersome listing processes, review of potential listings by political appointees and underfunding are blamed.²⁵ As the Centre for Biological Diversity states on its website:²⁶

²³ <https://www.fws.gov/project/species-status-assessment>

²⁴ Congressional Research Service, 2021, 12

²⁵ See Greenwald et al, 2019

²⁶ https://act.biologicaldiversity.org/A026WDSr9EShr5SvrXUWg2?sourceid=1008066&oa_ad_referrer=ad_action-alerts&utm_source=ad&utm_medium=website&utm_campaign=action-alerts

“The Fish and Wildlife Service has a convoluted, slow process for deciding whether to protect species: Though the law says those judgment calls just need to be based on solid science, 20-plus bureaucrats get involved in every decision.

Many of them have little knowledge of the species in question. Yet, simply because they're high up the management chain, they're able to veto protection at will — forcing agency scientists to deny safeguards that are desperately needed and, in turn, prompting us to challenge those denials in court.”

One interviewee advised that the federal agencies now only list species when they are sued for not doing so. Most of the Centre for Biological Diversity’s lawsuits are about listing decisions because they are, according to an interviewee from the Centre, the easiest to win.

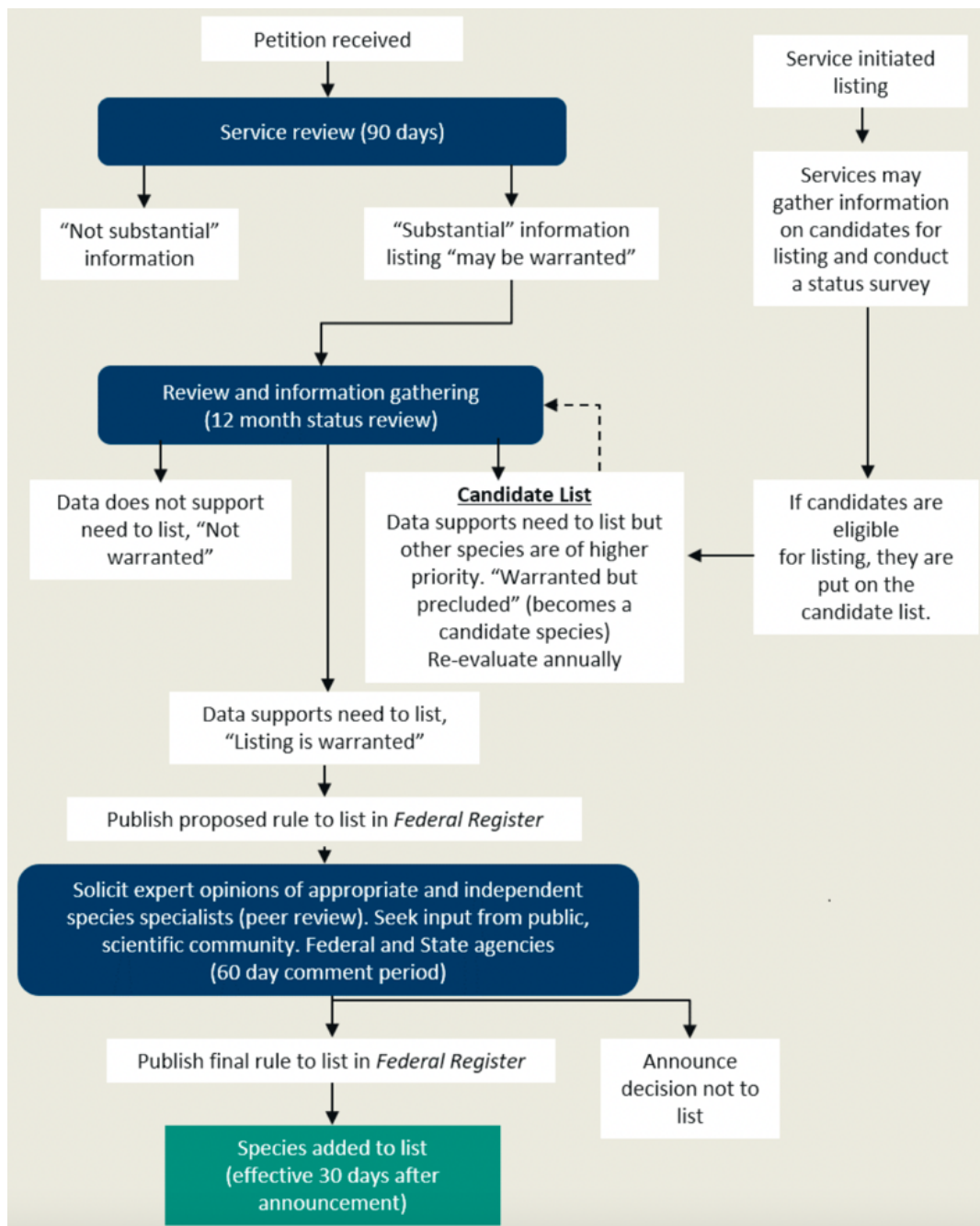


Figure 3. Process for listing species under the Endangered Species Act 1973²⁷

Listing process in Canada

Under SARA, listing decisions are also politically influenced because the Government, not COSEWIC, makes the final decision.

COSEWIC sits outside Government and is required to carry out its functions based on the best available information on the biological status of a species, including scientific knowledge, community knowledge and aboriginal traditional knowledge. By all accounts COSEWIC does this well, excluding economic considerations from its recommendations (although politics appears to have crept into COSEWIC committee appointments, with several nominees being rejected by Ministers without explanation).

COSEWIC does not, however, decide whether a species is listed. This is where, as described by one interviewee, the “wheels start to come off”. Instead, the listing decision is made by the Governor in Council on the Minister’s recommendation. In making his or her recommendation, the Minister must take into account (but does not have to accept) COSEWIC’s assessment.

The process goes like this:

- Within 90 days of receipt of a COSEWIC report the Minister must publicly notify a ‘response statement’ setting out how the Minister intends to respond to the assessment.
- The Minister forwards COSEWIC’s assessments to the Governor in Council.
- The Minister provides a recommendation on listing.
- Within 90 days of receipt of an assessment, the Governor in Council must either list the species, decide not to list the species, or refer it back for further information or consideration.

Before recommending to the Governor in Council that it list a species, the Minister must first undertake a series of consultations. The consultations process involves a cost-benefit analysis, which were described by interviewees as being poor because they do not quantify benefits well and because they are not peer reviewed.

The listing process does not set a timeframe within which the Minister must present COSEWIC’s assessment to the Governor in Council. This is a significant issue as it has allowed the Minister to undertake extensive consultation periods after receipt of a COSEWIC recommendation.²⁸ Delays in the process undermine COSEWIC’s ability to make listing recommendations based on the most up-to-date information. The greater the delay, the more chance that new evidence will become available, increasing the chance of an assessment being referred back to COSEWIC, and prolonging the process even further.

SARA only applies to federal land (except for emergency orders which are rarely implemented). The provinces and territories deal with listing decisions differently. Law in the Northwest Territories in Canada operates in a similar way to the federal Act, with a committee that conducts a biological assessment. However, unlike the federal process, the final listing decision is made by consensus among the members of the Conference of Management Authorities, which comprises different Government representatives and wildlife management boards.²⁹

²⁷ See Congressional Research Service, 2021

²⁸ Hutchings et al, 2016, 233-254

²⁹ Species at Risk (NWT) Act 2009

Among all the provinces and territories, the most scientifically robust listing process was that used in Ontario. When first enacted, Ontario's Endangered Species Act 2007 (Ontario's ESA) automatically listed species in accordance with scientific assessments. This completely removed Ministerial discretion in listing decisions. Adopting Ontario's approach has been heralded as a way in which SARA's listing process could be improved (see below *Spotlight: Transforming SARA by providing for automatic listing*). Unfortunately, the protective implications of Ontario's automatic listing have been undermined through the issuance of extensive exemptions and permits (see below *Spotlight: Ontario's automatic processes and subsequent weakening*).

Spotlight: Ontario's automatic processes and subsequent weakening

As noted above, SARA only applies to lands under federal jurisdiction. Since most of Canada's land is under provincial jurisdiction species protection often falls under provincial legislation. Of the seven sub-federal species at-risk laws in Canada, Ontario's ESA was considered the best, primarily because a species classified as Threatened or Endangered received automatic listing and associated protections.³⁰

Under Ontario's ESA, the Committee on the Status of Species at Risk is responsible for assessing and classifying species as either Extinct, Extirpated, Endangered, Threatened or of Special Concern. Upon classification by that Committee, regulations used to be automatically updated. There was no intermediate step of a Minister or third party having to confirm the listing.

Further, as soon as a species was listed it was automatically protected from harm or harassment and afforded general habitat protection. General habitat areas are areas on which the species depends, directly or indirectly, to carry out its life processes. After a recovery strategy is developed, a specific habitat protection is developed which replaces the general habitat protection.

In contrast, under SARA, prohibited acts only apply once the Governor in Council has confirmed COSEWIC's threat classification recommendation. Designation of critical habitat can be delayed further and need only be protected within 180 days of that area being identified in a recovery strategy or action plan. Formulation of recovery strategies and plans are mandatory but can occur years after a listing. For example, a recovery plan can take one year from listing for an endangered species, and two years for a threatened species. This leaves years between when a species is listed and when its habitat is protected - if that occurs at all. Missing action plans are a systemic issue under SARA.³¹ As of February 2021, of the 588 species listed under SARA only 345 had recovery strategies and 83 had action plans. In 2015, less than 12 percent of listed species had fully identified critical habitats.

Ontario's automatic listing and protections proved controversial, with business groups lobbying the Ontario Government to remove them from the Act.³² The processes were watered down by Ontario's More Homes, More Choice Act 2019, which suspends protections after a new species is classified.³³ One apparent rationale for the new post-listing delay is scientific uncertainty as to the status of a species. Experts argue that automatic protections should be restored and that the Government should turn to the precautionary principle outlined in Ontario's ESA in support of them; "where there is a threat of significant reduction or loss of biological diversity, lack of full scientific

³⁰ Bergman et al, 2020, 423–431

³¹ Palm et al, 2020, 1-13

³² Turcotte et al, 2021, 1474–1494

³³ Bergman et al, 2020, 423–431

certainty should not be used as a reason for postponing measures to avoid or minimize such a threat”.³⁴ The difference in protection timing is illustrated in Figure 4.

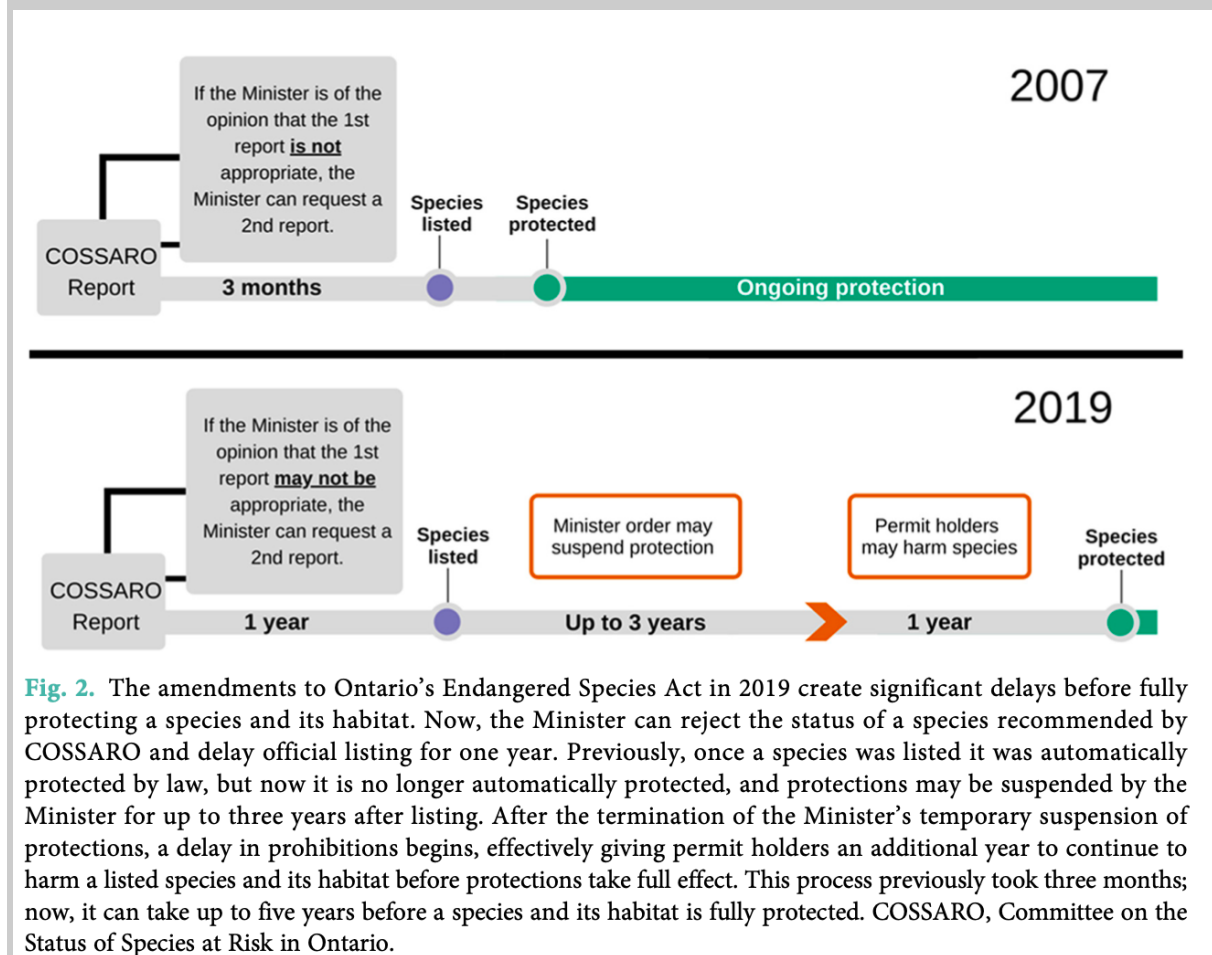


Fig. 2. The amendments to Ontario’s Endangered Species Act in 2019 create significant delays before fully protecting a species and its habitat. Now, the Minister can reject the status of a species recommended by COSSARO and delay official listing for one year. Previously, once a species was listed it was automatically protected by law, but now it is no longer automatically protected, and protections may be suspended by the Minister for up to three years after listing. After the termination of the Minister’s temporary suspension of protections, a delay in prohibitions begins, effectively giving permit holders an additional year to continue to harm a listed species and its habitat before protections take full effect. This process previously took three months; now, it can take up to five years before a species and its habitat is fully protected. COSSARO, Committee on the Status of Species at Risk in Ontario.

Figure 4. Difference in timing of protections pre and post amendments to Ontario’s Endangered Species Act 2007³⁵

Spotlight: Transforming SARA by providing for automatic listing

In 2021, ten academics from Canadian universities wrote a paper called *Fixing the Canadian Species at Risk Act: identifying major issues and recommendations for increasing accountability and efficiency*. The paper introduces SARA by saying that it was a promising piece of legislation because it allowed for scientific and economic input, deadlines for scientific-based recovery strategies, and protections for listed species on federally managed lands. However, since its enactment there have been few successes in species protection and recovery. Populations of listed species continued to decline 28 percent on average the following decade.

The paper identifies seven main problems and proposes solutions, both policy and legislative, to improve its implementation. Greater automation is a key theme of the paper’s proposed solutions, so that at certain trigger points actions occur without the need for further analysis or determination (e.g., an automatic listing process). The rationale being that automatic listings remove delays and are transparent.

³⁴ Bergman et al, 2020, 423–431

³⁵ Bergman et al, 2020, 423–431

If automatic listing proved too politically challenging, the paper suggests an alternative approach of automatic prohibitions (i.e., no additional activities) following listing of a species. The paper's proposed new listing process is set out in Figure 5.

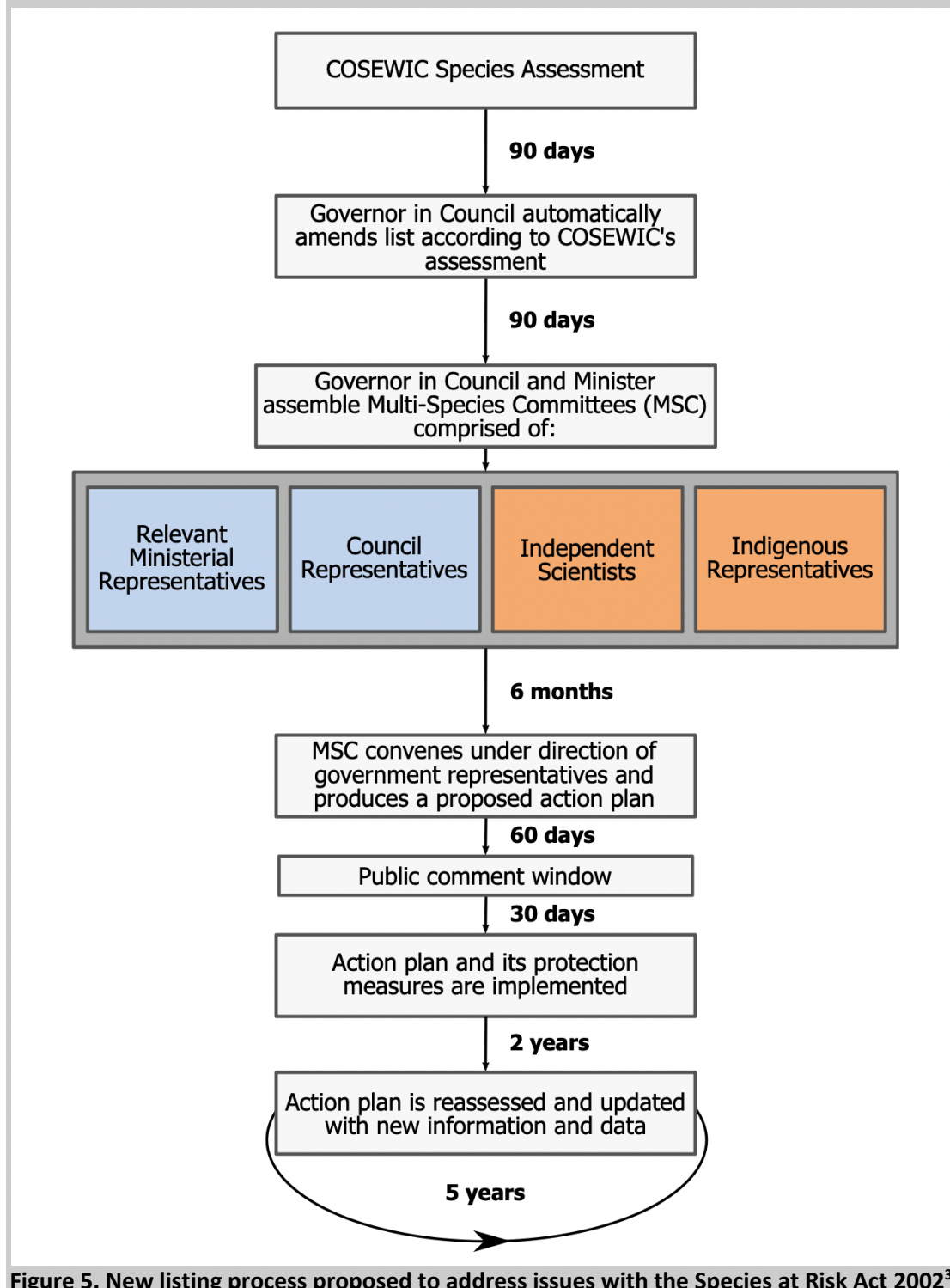


Figure 5. New listing process proposed to address issues with the Species at Risk Act 2002³⁶

Listing process in Australia

³⁶ See Turcotte et al, 2021, 1474-194, 1480

Australia's independent TSSC makes recommendations on whether a species should be listed under the EPBC Act. The Committee's assessments are not peer reviewed, but public input is sought. The final listing decision is made by the Minister after considering the Committee's advice.

The TSSC has recommended that the EPBC Act be amended to give the Committee listing authority, on the basis that listing is a scientific process and it would be more administratively efficient. The Committee accepts that the Minister should retain decision-making in relation to planning instruments, such as recovery plans and threat abatement plans, which may include other non-scientific considerations and statements about the Government's intended response.

Multi-species threat assessments

The U.S. system is not functioning well with a backlog of hundreds of species waiting for listing decisions or designation of critical habitat.³⁷ Similarly, COSEWIC has a backlog of species that may warrant listing, including over 500 vascular plants on the candidate list. In 2021, the average contract value to prepare a new extinction risk assessment for vascular plants was \$11,200,³⁸ putting the cost for assessing all 500 plants at \$5.6million CAD.

The US Fish and Wildlife Service has sought to reduce workload by making some plans multi-species i.e., habitat conservation plans and recovery plans, but this approach has not extended to Species Status Assessments, which are only allowed to focus on one species. SARA specifically provides for a multi-species or an ecosystem approach when preparing recovery strategies and management plans for species of special concern, but not for extinction risk assessments.

Multi-species extinction risk assessments could bring significant efficiencies to the listing process, especially when the species are sympatric and share the same threats. Concurrently assessing species from within the same ecosystem could enable better information gathering and understanding of threats. Wherever possible, COSEWIC will assess a bundle of related species i.e., a plant and its insect. Australia's listing of threatened ecological ecosystems (discussed below) compliments a broader assessment of species.

Listing commercially valuable species or bycatch

The protection and recovery of a threatened species is especially fraught when that species is also a valuable commodity, or the bycatch of a valuable industry. In most cases, to avoid triggering protections under threatened species law, the species is simply not assessed in the first place. If it is assessed, it is typically placed in a threat category with no regulatory impact or is exempted from regulations. Commercial fish and bycatch species are caught in this fray.

In Canada, SARA bestows broad powers to protect aquatic threatened species (both freshwater and marine). However, marine fishes comprise the species group least likely to be listed under the Act.³⁹ This is largely because marine wildlife are the subject of extensive commercial harvesting.

As of 2016, of the 32 occasions that the federal Government has had to list an endangered or threatened marine fish, only two have made the cut, both sharks with no current or prospective commercial value.⁴⁰ The lack of acceptance of COSEWIC's listing advice on marine fishes has been described as a 'statutory wound that has been allowed to fester'.⁴¹

³⁷ Noss et al, 2021, 1715-1724, 1717

³⁸ Kraus et al, 2021, 1088-1127

³⁹ Hutchings et al, 2016, 233-254

⁴⁰ Hutchings et al, 2016, 233-254

⁴¹ Hutchings et al, 2016, 233-254

Instead, the Government contends that listing is not required because all the necessary tools for recovery can be effected under the auspices of the Fisheries Act. That contention was rejected by the Court in relation to whether the Fisheries Act or the Canadian Environmental Assessment Act met the legal protection requirements of SARA in relation to protection of the Southern Resident Killer Whale's critical habitat.⁴²

In Australia, the tension is evident in relation to (among others) the Southern Bluefin Tuna, which is listed as Critically Endangered by the IUCN, but only classified as Conservation Dependent under the EPBC Act. Under the EPBC Act, a commercial fish species must be included on a list of exempt native specimens before it can be sold overseas. The Southern Bluefin Tuna is on that list. The Australian government maintains that the best prospect for safeguarding this highly migratory species is through continued exploitation by Australian fishers as this gives Australia a seat at the Commission for the Conservation of Southern Bluefin Tuna.⁴³

The interface between Aotearoa's Wildlife Act 1953 and fisheries law is discussed further in Appendix D. The predominant approach here and abroad has been to rely on fisheries legislation to manage fish and bycatch populations. But fisheries laws are inherently about maximising take, not protecting species, and they do not deal well with recovery.

The IUCN Red List system produces species extinction risk assessments that align well with fisheries management metrics (i.e., the IUCN assessments do not exaggerate threat status) so, for countries like Canada and Australia that use the IUCN criteria to assess the extinction risk of species, there should be no impediment to accepting scientific listing advice for fish.⁴⁴

Spotlight: The conservation dependent scalloped hammerhead shark discusses the TSSC's recommended approach for dealing with threatened marine species. There are likely other ways to deal with the interface between threatened species laws and fisheries laws, but we consider it important that species are listed in a threat category that reflects their extinction risk, and that there is some form of trigger which shifts regulatory control from fisheries laws to threatened species laws when fisheries management fails.

Finally, as for any other listed species, an exemption or permitting regime for a commercially valuable species should be carefully designed to not undermine protections afforded. For example, SARA enables fishing interests to supersede protections afforded to listed species by authorising prohibited acts through permits issued under other Acts. In 2014, pursuant to the Fisheries Act, Fisheries and Oceans Canada issued ten fishing licences relating to listed species.⁴⁵ In Aotearoa, the Wildlife Act provides a complete defence for the death, injury or possession of fisheries bycatch species (see Appendix D for discussion).

Issues with listing terrestrial species also arise, not when the species is commercially valuable, but when its habitat is. Listing species that find refuge in novel artificial habitats in intensively used landscapes has been a challenge in Aotearoa, e.g., some species of skink in Canterbury. Again, the issue is one of vested interests interfering in the protection of threatened species.

Lessons learnt: The listing process

⁴² Georgia Strait Alliance v Canada (Minister of Fisheries and Oceans), 2009 FC 1233

⁴³ Hutchings et al, 2016, 233-254

⁴⁴ Hutchings et al, 2016, 233-254

⁴⁵ Hutchings et al, 2016, 233-254

1. To effectively prevent extinctions, and to have credibility, threatened species lists should comprehensively address all species and accurately reflect species' extinction risk.
2. Extinction risk assessments should be independent from politics and vested interests.
3. Automatically listing a species according to its extinction risk assessment reduces opportunity for outside influence in the process.
4. Upon listing, species and their habitats can be immediately protected.
5. Multi-species extinction risk assessments may be a more cost-effective way of assessing species.

The benefit of being a listed species

All the jurisdictions reviewed include two forms of protections for species listed in the threat categories discussed above:

- (a) Prohibited acts - whereby certain acts in relation to the species are prohibited i.e., the species is protected via a no 'take' regulation; and
- (b) Habitat protection - whereby the species' 'critical habitat' cannot be destroyed or modified without permission.

Protection can be implemented in two main ways: via secondary legislation (i.e., the United States) or amendment to primary legislation (i.e., the United Kingdom and the EU).

Listing a species also triggers recovery planning for the species and its habitat. It also stimulates funding for its protection and recovery. Money allocated is almost always grossly inadequate and underfunding is a significant barrier to threatened species conservation.

We address prohibited acts, designation of critical habitat and recovery planning in detail the following sections, but first provide some general comment on the benefit of being listed under each jurisdiction reviewed.

Striking the right balance

If extinctions are to be prevented, protections are required in threatened species laws. The regime must, however, have a measure of flexibility and allow for some exemptions. To do otherwise risks perverse outcomes, including disincentivising the listing of species in the first place and the illegal destruction of species and/or their habitat. It also stymies the ability to undertake conservation actions that may necessitate disturbance of species or their habitats.

How to strike the right balance between protections and flexibility was discussed with many interviewees. Unsurprisingly, no one had the golden answer, but all indicated that a stick and a carrot were essential. We discuss enforcement and incentives separately below.

Benefit of being listed under the ESA

The ESA is one of the world's most powerful biodiversity conservation laws. Its regulatory reach onto private property is astonishing given the strength of private property rights in the U.S. The true power of the Act is often told through the story of an unassuming but once endangered 3-inch fish in Tennessee (see below *Spotlight: How a tiny fish stopped a dam*).

Spotlight: How a tiny fish stopped a dam

In 1978, the U.S. Supreme Court issued a decision on the ESA that changed the course of species protection for decades to come. In just three small words, “whatever the cost”, the Court set a precedent for species protection under the Act that endures to this day.

The case involved an application for a dam on the Little Tennessee River that would improve flood control, store water for irrigation and create a large real estate development. Unfortunately for the applicant, the proposed dam location coincided with the only known habitat of a small, plain looking, tan coloured, bottom dwelling river fish, called the snail darter (named after its favourite food and its method of travel). Uncharismatic as it may be, the fish was listed as Endangered in 1975, signaling the start of a David and Goliath battle. Arguments were made that building the dam would be a violation of the ESA.

In the first round of decisions, the fish lost, with the trial Judge asking counsel mid-hearing “Do you really think that Congress would want me to stop an important project for just any endangered creature, for some red-eyed cricket?”. Round two went the darter’s way. A law clerk with the Court of Appeals later told counsel that during the hearing one of the Judges was not taking notes, but rather composing a limerick:

*“Who can surpass the snail darter?
The fish that would not be a martyr.
It stymied the dam,
Near the place where it swam.
Can you think of a fish any smarter?”*

The third and final round in the U.S. Supreme Court was hard fought but, in a six-to-three vote, the Court determined that the dam would violate the Act. The Court held “the plain intent of Congress in enacting the Act was to halt and reverse the trend toward species extinction, whatever the cost”. No environmental law in the U.S. or any other nation has ever been interpreted to have such a meaning. In three short words the Court gave the prevention of species incalculable value.

Bending under pressure from the fallout of the case, Congress amended the ESA to allow for exemptions to be determined by an Endangered Species Committee, nicknamed ‘The God Squad’ by the media. Remarkably, the dam was not granted one.

In the end, the dam was given the green light via law unrelated to the ESA.⁴⁶ Fear not for the snail darter though, in a final twist the fish had a separate population secretly residing in another creek. Since then, relocated populations have also taken up happy residence in two other rivers. In 2021, to reflect individual fish numbers, the snail darter was removed from the endangered species list due to recovery.

The ESA provides several protections to listed species. Endangered fish and wildlife (not endangered plants or threatened species) are protected through prohibited acts. The critical habitat of both endangered and threatened species is designated. Recovery plans must also be promulgated for both endangered species and threatened species.

Significantly, for endangered and threatened species, the Act includes a mechanism to ensure that the federal Government itself does not contribute to listed species decline. The rationale being, if the ESA is going to prevent certain actions on private land, then the federal Government must also

⁴⁶ A ‘rider’ was slipped into a multi-billion-dollar appropriations bill which granted the dam company an exemption. A rider is an undemocratic tactic by which a completely unrelated but controversial provision is appended to a proposed law that enjoys wide popularity.

ensure that it is not causing harm. It cannot be a hypocrite. The mechanism is termed the 'no jeopardy' rule and has significant ramifications for species protection in the U.S.

Under the no jeopardy rule, all federal agencies must ensure that any action authorised, funded or carried out does not jeopardise the continued existence of a listed species or result in the destruction or adverse modification of the species' critical habitat. The best available science must be used when making that assessment.

When considering the words 'authorised, funded or carried out', the scope of federal actions captured by the rule is extensive. Incidental take permits, safe harbour agreements and CCAAs (all discussed further below) trigger the jeopardy assessment. But the rule has wider application. For example, pesticide authorisations by the Environmental Protection Agency must assess jeopardy as pesticides are a known threat to many bird and invertebrate species, aviation administrative decisions must assess jeopardy because helicopters and low-flying planes can affect tree-nesting birds, fish catch limits must assess jeopardy because of bycatch issues. Issuing leases to graze cows on federal land triggers the rule. As do permissions to drain wetlands. The list is exhaustive.

If a federal action has the potential to jeopardise a species, a series of consultations and biological assessments are required. A decision is then made as to whether jeopardy or adverse modification might occur and, if it was to occur, what alternatives could be implemented and what mitigation applied.

An exemption can be granted to a federal agency to undertake an action that would result in jeopardy. Exemptions are determined by a formidable Committee comprised of the Secretary of Agriculture, the Secretary of the Army, the Chairman of the Council of Economic Advisors, the Administrator of the Environmental Protection Agency, the Secretary of the Interior, the Administrator of the National Oceanic and Atmospheric Administration, and an individual from each State affected (aka The God Squad). Exemptions are limited to cases where there are no reasonable and prudent alternatives to the agency action, where the benefits of the action clearly outweigh the benefits of alternatives consistent with conserving the species or its critical habitat and such action is in the public interest, and the action is of regional or national significance. The God Squad has only been asked to grant an exemption six times in the last four decades, only one has been issued.

Of the more than 6,800 formal interagency consultations that took place between 2008-2015 only two projects resulted in the conclusion that the proposed agency action would likely place a species in jeopardy.⁴⁷ A U.S. Fish and Wildlife interviewee advised that those figures are representative of the Agency's keen desire to avoid jeopardy wherever possible. Others advised jeopardy is almost never found because the federal agencies are only considering jeopardy on a site-by-site basis. At the individual property scale, habitat loss may not trigger jeopardy, but these actions are having a significant cumulative effect on the destruction of critical habitat. This approach is undermining the effort invested in getting the critical habitat designated in the first place.

Finally, the ESA allows for 'citizen suits' so that any person (with standing) may commence a civil proceeding against any other person, including Government agencies, who are alleged to be in violation of the Act. Citizen suits are hugely important in the U.S. system and play a significant enforcement role under the ESA. In the 1990s, successive and successful citizen suits against federal agencies in relation to not listing the northern spotted owl, not designating all of its critical habitat, and issuing exemptions to forestry companies to cut trees within its critical habitat, resulted in cancellation of timber sales and protection of 10 million acres of old growth forest for the owl.

⁴⁷ Holmer, 2016

Citizen suits have been so effective that avoiding them is often the most powerful ‘stick’ under the ESA.

Combined, the above mechanisms establish a robust framework for the protection and recovery of threatened species. The Act has several deficiencies, discussed throughout this report, and its implementing policies (secondary law) have been weakened over time to accommodate sector interests, landowners and political interests. But, overall, it is a sound place to start when attempting to frame threatened species law because it is both principled and conceptually robust.

Effectiveness of the ESA

Since enactment, more than 2,400 species have been listed as endangered or threatened under the ESA. There are currently approximately 1,600 species listed, 500 of which live in Hawaii (‘the endangered species capital of the world’). The Act is reported to have prevented the extinction of roughly 291 species and has been particularly effective for marine mammals and turtles that were threatened due to over-exploitation. It is commonly quoted that to date it has saved more than 99 percent of species under its protection, meaning that once a species is listed, its chance of going extinct is very low.

Having said that, over half (52 percent) of listed species continue to decline and only a few species have been down or de-listed. Of those 91 species delisted:⁴⁸

- Approximately 65 percent were delisted due to recovery.
- Approximately 15 percent were delisted due to new information, changes in the law, or improved scientific understanding of the species.
- Approximately 12 percent were deemed extinct.
- Approximately percent were delisted due to taxonomic reclassification of the species.

Stakeholders use these data as evidence that the Act is working (low number of extinctions of listed species) and conversely that it is ineffective (low number of delisted species due to recovery). The U.S. Fish and Wildlife Service acknowledge that the Act is seen as one of the most controversial laws with issues being raised around:⁴⁹

- The Act’s effects on private property and landowners.
- The inability to conserve species before it is necessary to list them.
- The cost of listing species and the resulting economic impacts.
- The availability of funding for the Act.
- Incentives for conservation under the Act.
- States’ role in conserving listed species.
- Delays in listing, delisting, and reclassification of species.
- Litigation related to the Act.

Benefit of being listed under SARA

Prior to introduction of SARA, Canada was where Aotearoa is now. It had a Wildlife Act and a scientific committee which assigned a threat status to species, but it had no specific threatened species law.

⁴⁸ Congressional Research Service, 2021, 52

⁴⁹ Congressional Research Service, 2021, 53

Now, under SARA, Canadian federal protections afforded to listed species are analogous to the U.S. in terms of general prohibitions, identification of critical habitat and recovery actions. Its terrestrial application is severely curtailed however, because SARA only applies to federal lands, of which there is very little. In all other areas, wildlife protection is left to individual Provinces and Territories, among which endangered species law is often deficient or absent. Where the federal law has greatest impact is in the marine environment. Almost all fish and marine species are captured by SARA.

Benefit of being listed under EPBC Act

In Australia, listed species are protected via prohibited acts and the designation of critical habitat. Species are also subject to recovery planning.

Further, as noted above, some listed species and ecosystems are matters of national environmental significance under the EPBC Act. A person must not take an action that has, will have, or is likely to have, a significant impact on matters of national environmental significance without approval from the Minister. For example, listing under the EPCB Act was critical to ensuring that major economic development did not compromise the habitat of the endangered pygmy bluetongue lizard in Australia.⁵⁰

Australia's biodiversity crisis

Australia has one of the world's worst records for extinction and lack of protection for threatened species. It has experienced the largest documented decline in biodiversity of any continent over the past 200 years and has one of the highest human-driven extinction rates of land mammals (50 percent of global mammal extinctions in the last 200 years have occurred in Australia). It is rated second (after Indonesia) in the world for ongoing biodiversity loss.

Australia's track record with threatened species has come under intense scrutiny in recent years. In 2018, the Senate referred a review of 'Australia's faunal extinction crisis' to the Environment and Communications References Committee for inquiry. The report was due end of 2018 but the Committee has been granted multiple extensions. At the time of writing, a final report has still not been issued.

In an interim report prepared by the Committee, Australia's EPBC Act was heavily criticised by submitters. The Committee concluded that there have been significant failures in its implementation, including the use and resourcing of compliance and protection mechanisms.⁵¹ Issues which were also reflected in a 2020 study which stated that Australia's decline in biodiversity is likely the result of both delayed ecological responses to conservation actions and ineffective approaches.⁵²

The Committee recommended a new legislative approach and that an independent Environmental Protection Agency was required to oversee compliance of the new Act.⁵³ Labor Senators supported the recommendations.

⁵⁰ Garnett et al, 2018, 108

⁵¹

https://www.aph.gov.au/Parliamentary_Business/Committees/Senate/Environment_and_Communications/Faunalextinction/Interim_report/c04

⁵² Watts et al, 2020, 304–311.

⁵³

https://www.aph.gov.au/Parliamentary_Business/Committees/Senate/Environment_and_Communications/Faunalextinction/Interim_report/c04

Key to Australia's issues appears to be the EPBC Act's stance that merely listing a species and affording it protection via prohibited acts is enough. Recovery planning and critical habitat designation are not mandatory in Australia. But, as is demonstrated by Australia's dire biodiversity statistics, listing alone cannot be assumed sufficient to maintain a species. Supporting mechanisms are also required.

Threatened Species Commissioner

In 2014, to complement the Government's responsibilities for threatened species protection and recovery under the EPBC Act Australia appointed a Threatened Species Commissioner. The Commissioner, and associated Office of the Threatened Species Commissioner, sit within the Ministry for the Environment. The Commissioner's role is to champion the development and implementation of practical on-the-ground actions to protect and recover threatened species, including implementing Australia's Threatened Species Strategy and providing policy advice on threatened species investments.

It is a non-statutory advisory role only; the Commissioner is not able to prevent decisions and the Government is not obliged to accept any recommendations from the Commissioner. Critics argue that the Commissioner is simply another layer of ineffective bureaucracy.

Lessons learnt: The benefit of being a listed species

1. Listed species are provided protections, including habitat protection, and are subject to recovery planning.
2. A combination of mechanisms is required to protect and recover threatened species, simply preventing take is not enough.
3. The regulatory implications of listing should reflect threat statuses and should connect with planning laws.
4. It is important that the Government lead by example in protecting threatened species by ensuring that its own actions do not put such species at risk.
5. Current threatened species laws are still enabling biodiversity loss.

Prohibited acts and exemptions

Once listed, a species is afforded mandatory protections via the prohibition of certain acts impacting that species. Table 1 summarises those acts for each of the jurisdictions reviewed.

The no take provision in the ESA is very powerful, given the breadth of actions covered by the term (see definition in Table 1) and how it has been interpreted e.g., take has been successfully inferred using satellite imagery of habitat clearance. However, it only applies to endangered species, and excludes plants (see below *Spotlight: What about the plants?*). Similar no kill or injure provisions in Canada and Australia apply more broadly, to threatened and even vulnerable species (in Australia), and to animals and plants.

Jurisdiction	Prohibited act	Threat category	Species	Definitions
U.S. Endangered Species Act	Import, export, take, possess, sell, deliver, receive, carry, transport, ship	Endangered	Animal, fish, bird, amphibian, reptile, mollusk, crustacean, arthropod, invertebrate	Take means to harass, harm, pursue, hunt, shoot, wound, kill, trap, capture, or collect, or to attempt to

	Import, export Deliver, receive, carry, transport or ship, sell or offer to sell Under federal jurisdiction only: a) Remove and reduce to possession b) Maliciously damage or destroy Outside federal lands: Remove, cut, dig up, damage or destroy in knowing violation of any State laws	Endangered	Plant	engage in any such conduct c.f. the Wildlife Act 1953's definition of take includes taking, catching, or pursuing by any means or device, and also includes the attempt to take
Canada's Species at Risk Act	Kill, harm, harass, capture or take Possess, collect, buy, sell or trade	Extirpated, endangered or threatened Extirpated, endangered or threatened	Wild animal, plant or other organism Wild animal, plant or other organism	
Australia's EPBC Act	Kill or injure Take, trade, keep or move	Critically endangered, endangered, threatened, vulnerable	Species or ecosystem	

Table 1. Prohibited acts

Spotlight: What about the plants?

The ESA's no take rule has far-reaching application for fish and wildlife. Under the law, it is an offence to take listed endangered fish or wildlife within the U.S. or the territorial sea. That includes on private land.

The same cannot be said for endangered plants, which are only protected on federal lands. On federal land it is an offence to remove and have possession of any endangered plant or to maliciously damage or destroy any such species. Outside of federal land, plants are only protected if a person removes, cuts, digs up, damages or destroys an endangered plant in knowing violation of any State law or regulation or in the course of a State criminal trespass law.

Plants are captured by take provisions in Canadian and Australian laws. In these jurisdictions, even threatened (not just endangered) plants cannot be killed, injured, harmed, kept, or traded (see Table 1 for full list of prohibited acts for each jurisdiction).

Even though the U.S. no take rule does not apply to plants, the impact of an incidental take on a listed plant species is assessed under the jeopardy provision of the ESA, i.e., before issuing an incidental take permit the U.S. Fish and Wildlife Service must determine that the take will not jeopardise the continued existence of a listed plant or result in the destruction or adverse modification of its critical habitat.

Prohibited acts set an immediate baseline of permissible interactions with listed species. The restrictive effect of prohibited acts on private development is the most controversial aspect of threatened species legislation. It has resulted in perverse incentives such as ‘midnight bulldozing’ of the habitat of pre-listed species and a ‘shoot, shovel and shut up’ mentality.

To appease landowners and industry, threatened species laws have developed exemptions to prohibited acts for those wishing to develop lands that harbour listed species. There are several mechanisms by which this can occur, and each of the jurisdictions reviewed take a slightly different approach.

When developing an exemption framework threatened species law must be careful to provide just the right amount of flexibility. Too much flexibility, and the conservation objectives of the Act are undermined. Prohibited acts mean nothing if an exemption is too readily obtained (see below *Spotlight: The flip side of Ontario’s automatic protections*). Not enough flexibility, and perverse outcomes abound. It is a difficult balance to strike, but a very important one to get right. Made easier if positive incentives are also part of the package.

Threatened species law must also walk a fine line of developing an exemptions process that is both rigorous, so that effects on species and habitats can be assessed, but not too burdensome as to discourage applicant involvement. Complacency or an unwillingness to engage with threatened species law is a serious problem. It can lead to widespread incremental harm to listed species and their habitats which is unmonitored and, consequently, unenforced. Encouraging applicant involvement in a permitting system can be achieved using incentives (a plethora of which exist), but a stick is also likely required.

Incidental take permits and habitat conservation plans in the U.S.

In the U.S. an incidental take permit can be obtained for the otherwise prohibited act of taking a listed fish or wildlife species. Agency permission for such takes is contingent upon approval of a habitat conservation plan. Habitat conservation plans emerged as a tool of compromise to manage species-development conflict in the 1970s. They allow landowners to take listed species so long as:

- The taking will be incidental;
- The applicant will, to the maximum extent practicable, minimise and mitigate the impacts of such taking;
- The applicant will ensure that adequate funding for the plan will be provided; and
- The taking will not appreciably reduce the likelihood of the survival and recovery of the species in the wild.

Habitat conservation plans benefit the landowner by allowing them to undertake a prohibited act. The permission *per se* is the benefit. Controversially, in 1998 a further benefit was added, a

guarantee that so long as a habitat conservation plan was properly implemented no additional requirements or restrictions would be imposed on the landowner during the duration of the plan. The 'no surprises' policy, as it is called, means that even if a species declined under a plan, a landowner would not be required to do anything more except in extraordinary circumstances.

The concept is astonishing given that habitat conservation plans in the U.S. are designed to enable the incidental take (death or injury) of the most imperiled species, those that are already endangered. When viewed in the context of climate change, and the largely unknown but dynamic way in which species will have to adapt over time, it is difficult to see how anything but a very short no surprises policy would ensure positive conservation outcomes for the subject species over the lifetime of the plan, some of which can last for 20 plus years.

Two decades after its creation, is it hard to envisage a no surprises policy being implemented now without legal challenge. Simply put by one preeminent U.S. lawyer who we interviewed; it is illegal. We do not consider that an analogous policy should be adopted in Aotearoa.

The very first habitat conservation plan was devised in the 1970s to enable development of an area inhabited by two listed butterflies. The plan preserved 90 percent of the butterflies' habitat while still allowing some construction. It was hailed as a model for managing species-development conflict and, in 1982, the concept was enacted into law. Habitat conservation plans were initially sparingly used, only 14 were created in the first decade. All that changed in 1998 when the no surprises policy was introduced. As of May 2022, 1,405 habitat conservation plans have been approved on non-federal land in the U.S., many of which cover multiple species and vast areas of land.⁵⁴ The habitat conservation handbook states that plans have achieved significant conservation for covered species and includes specific examples of that success. Unfortunately, not all plans have been effective, and some have been abandoned.⁵⁵ Biodiversity offsetting around the world is a similar story of broken promises.

Under the ESA, habitat conservation plans are meant to specify:

- The impact which will likely result from the taking;
- Measures to minimise and mitigate such impacts, and the funding that will be available to implement such measures; and
- Alternatives and why such alternatives are not being utilised.

The U.S. Fish and Wildlife Service has produced a handbook that provides further guidance on plan requirements. In 2000, the handbook was strengthened with a five-point policy focusing on biological goals and objectives, adaptive management, monitoring, permit duration and public participation. It has also been updated to recognise the need to integrate climate change effects into plans.

The adaptive management component of the five-point policy was introduced, in part, in response to criticism of the no surprises policy. As there are no assurances in nature, giving long term regulatory assurances without room for improvement or change was no longer seen as appropriate. Rather, landowners should adapt to unanticipated species management requirements. The policy is an improvement. However, its application is limited as the adaptive management required need only occur within the confines of the assurances given. A landowner is not required to undertake adaptive management measures that go beyond what was agreed at the outset.

⁵⁴ <https://ecos.fws.gov/ecp/%20report/conservation-plans-type-region>

⁵⁵ Holmer, 2016

The adaptive management approach was also introduced to deal with circumstances where the scientific information was either incomplete, or still evolving. In the absence of adequate scientific information about a species' biological status and habitat needs, the use of habitat conservation plans has raised concerns. Without sufficient scientific knowledge about a species, it is difficult to assess the impact of a take and what measures are required to minimise and mitigate that impact (see below *Spotlight: How many acres is a bat worth?*). In such circumstances, a precautionary approach should be adopted. In some cases, it might be appropriate to decline the incidental take altogether, remembering that these plans address take (the killing or injuring) of species already endangered or threatened.

The US Fish and Wildlife Service plays an active role in helping landowners develop plans, including financial support via habitat conservation planning assistance grants. It is much easier to impose mitigation measures prior to a development being built, so federal agencies try to engage with landowners and developers before they have proceeded too far through the planning and build stages.

Mitigation proposed in the plans can include permanent habitat protections that extend past the expiry of take permits i.e., land acquisition grants can be allocated by the U.S. Endangered Species Conservation Fund to States to purchase land in accordance with the terms of a habitat conservation plan.

Hawaiian habitat conservation plans

Hawaiian's State Endangered Species Statute HRS 195D loosely follows the federal ESA but contains some notable differences when it comes to habitat conservation plans. Overall, it is a more specific and rigorous legislative framework for such plans. Approval of habitat conservation plans is restricted to cases where the cumulative activities contemplated by the plan are environmentally beneficial and there is an overall net gain in the recovery of the species (as opposed to no net loss) (see below *Spotlight: Overall net gain*).

Key differences between the federal and Hawaiian state law for habitat conservation plans are:

- Under the ESA, the Secretary only needs to provide 'opportunity for public comment' before approving a habitat conservation plan. Hawaiian law is far more prescriptive when it comes to obtaining public input. A habitat conservation plan must be publicly notified and a public hearing held before it can be approved. Further, a plan can only be approved in cooperation with other government agencies, private organisations and landowners.
- A habitat conservation plan in Hawaii has a positive obligation to improve a species survival, whereas a federal plan need only ensure that a species' chance of survival does not get worse. Hawaiian law explicitly states that a habitat conservation plan must protect, maintain, restore or enhance species' habitat and increase the likelihood of recovery of any listed species the focus of the plan. The ESA merely requires that a plan not reduce the likelihood of the survival and recovery of the species.
- Habitat conservation plans in Hawaii must be based on the best available scientific and other reliable data. There is no analogous requirement in the ESA.
- Plans in Hawaii must identify steps that minimise and mitigate all negative effects (including cumulative) of the incidental take. Assessment of cumulative effects is not required in habitat conservation plans under the ESA.

- Under Hawaiian law, habitat conservation plans must provide reasonable certainty that the species' habitat will be maintained. No required for certainty exists under the ESA, indeed federal plans can be approved on the basis of "such other assurances" that the plan will be implemented.
- Hawaiian plans must contain objective, measurable goals which will contribute significantly to conservation efforts, time frames for achieving those goals, provisions for monitoring process against those goals and adaptive management strategies that identify actions to be taken if goals are not being met. The ESA includes no such requirements.

Some of the above Hawaiian provisions are addressed in the U.S. habitat conservation plan handbook, but that is not a statutory document and can be amended at any time.

Aspects of Hawaiian State law, such as overall net gain to the recovery of the species, that cumulative impacts of the species must result in net environmental benefits, and a positive obligation to improve the survival of the subject species, constitute, on paper at least, robust safeguards and protections for species proposed to be taken.

In practice, however, the system is showing cracks. For example, the Endangered Species Recovery Committee, which issues incidental take permits and approves habitat conservation plans, often struggles to define 'overall net gain' and 'net environmental benefit', especially in the context of mitigation measures that take a long time to achieve for species where there is insufficient scientific data. In some cases, mitigation has been accepted that does not directly improve the conservation outcome of the species being killed (such as research dollars).

One interviewee advised that the system was designed for a time of stability, and it is not equipped to deal with the multitude of threats currently facing Hawaii e.g., pest species, drought, intense flooding events and wildfire.

It is unclear whether habitat conservation plans in Hawaii are producing improved conservation outcomes for species. Notwithstanding several enquiries, we were unable to locate any reporting on their effectiveness. Highlighting the need for reporting requirements and greater accountability in threatened species laws at all levels, from the landowner to the decision-maker.

Spotlight: How many acres is a bat worth?

Habitat conservation plans require mitigation measures to address any negative effects of activities on listed species. Mitigation is a controversial topic, particularly in cases where there is insufficient scientific knowledge to fully understand what measures are required to offset the damage inflicted.

In Hawaii, mitigation has proved controversial in the context of incidental mortality of the hoary bat. The hoary bat is the only native terrestrial mammal in Hawaii and is listed as Endangered under federal and state law. It is particularly vulnerable to wind turbine strike, making wind power a significant threat to its continued survival. Incidental deaths of the hoary bat are expected to rise as more Hawaiian wind farms come online to help the State achieve its goal of being 100 percent renewable energy by 2045.

There are currently 11 habitat conservation plans in Hawaii and six of them relate to wind farms, all of which have approved incidental takes for the hoary bat. Development of habitat conservation plans for the bat has been problematic because much of the basic information on ecology and life

history of the species that is essential for designing such plans is limited or lacking. For example, there is no data to infer with statistical confidence the effect that conservation measures will have on the bat.

As a result of that scientific uncertainty, the scale and cost of bat mitigation in Hawaiian habitat conservation plans has changed over time as more biological and ecological information has become available. Initially, mitigation comprised financial contributions to research. Thus, incidental takes of the endangered bats were being allowed in exchange for money. ‘Blood money’, as one interviewee described it. Over time, mitigation has involved financial contributions to land acquisition for new habitat, and habitat restoration.

Between 2014 and 2017 several of the authorised wind projects exceeded approved take numbers of the hoary bat.⁵⁶ In 2019, Hawaii’s biggest wind farm sought to amend its habitat conservation plan to kill 160 more bats than the 60 already approved (the project had been operating in breach of its approved incidental take for over a year before applying to change its habitat conservation plan, having already killed more bats than first approved). In exchange for the additional bat takes, the proponent contributed \$2.75 million to a land purchase and proposed future habitat creation and restoration. A question then arose: many acres is a bat worth?⁵⁷ How much land does a wind farm operator have to protect, create or restore in order to offset the death of one bat? A particularly challenging question given the uncertain scientific knowledge about the bat’s biology and habitat requirements.

Ultimately, the amended habitat conservation plan was approved, enabling more bat takes. The issue is unparalleled among endangered species management in Hawaii and has led to development of the Hawaiian Hoary Bat Guidance Document 2019 (2nd Ed) to guide habitat conservation plans for the bat.

Incidental take of the hoary bat highlights the potential pitfalls of habitat conservation plans when there is scientific uncertainty. Given that habitat conservation plans provide for incidental takes of the most imperiled species, those at the brink of extinction, a science led, precautionary approach should be adopted - which presumably means that wind farms should be scaled back or avoided in some bat areas.

Canada’s exemptions

Under SARA, an exception to a prohibited act can be authorised in one of three circumstances:

- The activity is scientific research relating to the conservation of the species and conducted by qualified persons;
- The activity benefits the species or is required to enhance its chance of survival in the wild; or
- The activity incidentally affects the species.

An applicant applying for an exemption must satisfy three pre-conditions: all reasonable alternatives to the activity have been considered and the best solution has been adopted, all feasible measures will be taken to minimise the impact of the activity on the species, or its habitat, and the activity will not jeopardise the survival or recovery of the species.

⁵⁶ Hawaiian Hoary Bat Guidance for Wind Energy Projects, 2021, https://dlnr.hawaii.gov/wildlife/files/2021/01/Draft-Hawaiian-Hoary-Bat-Guidance-Document_011321.pdf

⁵⁷ Dawson T, 2019, ‘Wind farm barely gets preliminary OK to kill 160 additional endangered bats’, August 2019, <https://www.environment-hawaii.org/?p=11635>

The Act also provides for two activity-type exceptions. First, the prohibitions do not apply to activities permitted by a recovery strategy, an action plan or a management plan. Secondly, they do not apply to activities undertaken in accordance with conservation measures for the species under land claims agreements. The former could cover pest management which, in the short term, may harm a listed species but be beneficial for the wider ecosystem in the long-term.

Spotlight: The flip side of Ontario's automatic protections

Ontario's ESA prohibits acts against listed endangered or threatened species, including the damage or destruction of their habitat. As discussed in *Spotlight: Ontario's automatic processes and subsequent weakening*, prior to 2019, the Act's protections were imposed immediately upon listing. The automatic regime, modelled off the ESA, was a main reason why the province's legislation was seen as leading within Canada, and why adopting it has been proposed as way of strengthening the federal Act.

The effect of the Act's automatic protections has been significantly undermined by a plethora of exceptions that have been approved by way of agreement, permit or conditional exemption:

- Stewardship agreement
- Landscape agreement
- Health and safety permit
- Protection or recovery permit
- Overall benefit permit
- Social or economic benefit permit
- Aboriginal permit
- Conditional exemptions
- Other exemptions
- Exemptions provided via other laws

The number of exceptions granted significantly increased from 2013, when regulations were changed to allow persons to be conditionally exempted from the Act's prohibitions without the need for a permit (see Figure 6).

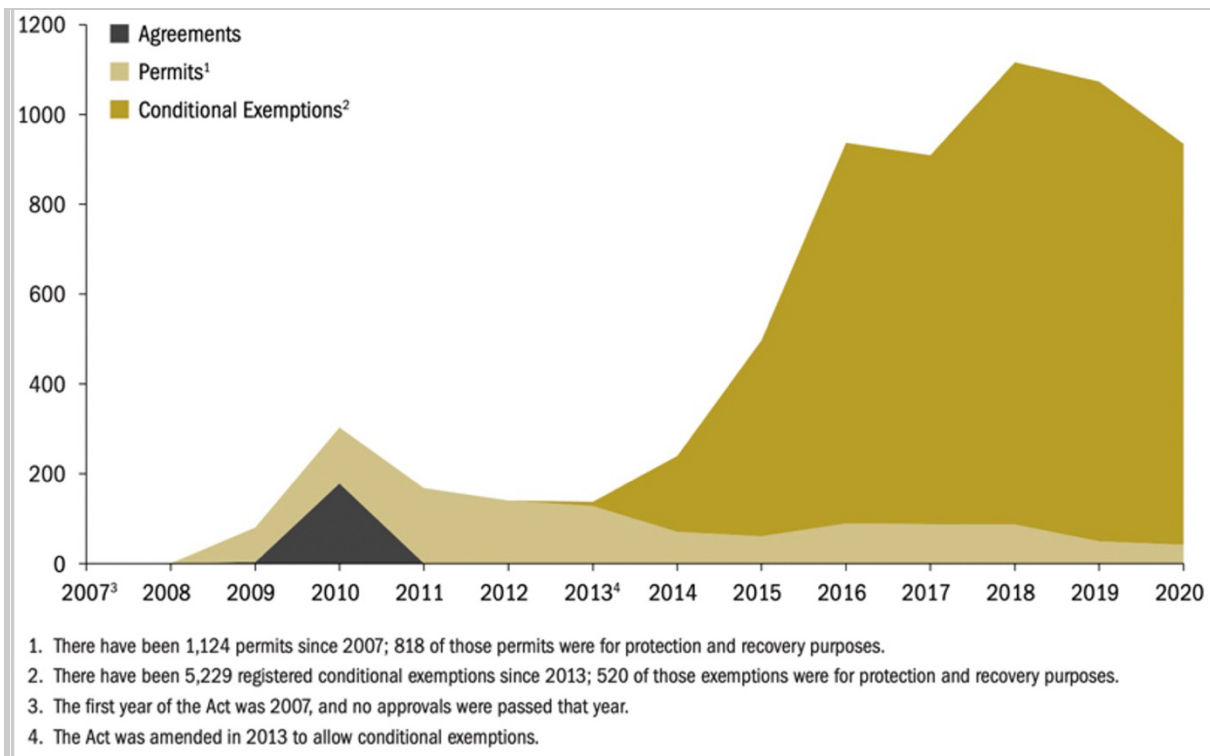


Figure 6. Approvals under Ontario's Endangered Species Act 2007, 2007-2020

The 2013 change was an attempt to streamline some approvals, particularly for wide ranging species that regularly triggered the Act such as eastern meadowlark and the bobolink. These are species that are listed not because of population numbers (which are reasonably high), but because of their rates of decline. The bobolink for example inhabits grasslands and farm fields. Mowing of hay during the breeding season can inadvertently kill or disturb nesting adults and young birds, and destroy eggs and nests, triggering the need for a permit.

Since its introduction in 2013, there have been 2,010 activities that have registered for conditional exemptions for bobolink and 1,964 for eastern meadowlark, ranging from wind facility operations to drainage works construction. More than 90 percent of the conditional exemption registrations for both of these birds do not concern imminent threats to human health or safety.

In effect, the conditional exemption regime has green-lighted harmful activities *en masse*. Those activities are not required to eliminate harm or provide an overall benefit to species. The conditions of exemption include things like registering with the Ministry, developing a mitigation plan, keeping activities a specified distance away from important habitat features, limiting the timing of activities, creating or enhancing habitat elsewhere, species monitoring and maintaining habitat features. But the system is entirely proponent driven and expects landowners to know about a listed species and to understand its needs.

One interviewee has recently been engaged by the federal government to study the exemptions regime for Blanding's turtle, which is a wide-ranging species the subject of conditional exemptions. The turtles have been impacted by 1,403 approvals since 2007, and the species has declined by more than 60 percent over the last three generations because of ongoing habitat loss in Ontario. The outcome of the study will inform the federal Government's decision whether to intervene under emergency powers of the SARA.

In 2020, Covid response legislation permanently exempted forestry on Crown lands from Ontario's ESA, despite federal Government warnings that the exemption did not conform with the federal Act. Nothing has been done on either side in response; the exemption still stands and the federal Government has not intervened.

In total, between 2008 (when Ontario's ESA came into force) and 2020, 6,262 approvals to impact listed species were granted, compared to only 59 approvals for protection and recovery.

In November 2021, the Office of the Auditor General of Ontario undertook an audit of Ontario's protection and recovery of species at risk. It found that:

- Approvals are not assessed for how they cumulatively affect species at risk and their habitats.
- Permit applications to harm listed species or their habitats are always granted.
- In 2020, 96 percent of approvals granted were conditional exemptions, which cannot be denied or tailored for specific circumstances.
- The Environment Ministry has not assessed the effectiveness of overall benefit permits that allow harmful activities but require that a species be made better off.
- The Environment Ministry has delayed issuing some permits for conservation work, while fast-tracking some permits for development.
- There is an increasing use of social or economic benefit permits that allow harmful activities but do not require that the species is made better off.
- The Environment Ministry does not charge fees for approvals, although this would discourage activities that harm species at risk and their habitats.

Ontario is a clear example of a jurisdiction that has not struck the right balance between protections and flexibility. It is interesting to compare Ontario's approach with the U.S. federal system. The laws of both jurisdictions impose (or at least did until recently in Ontario's case) automatic protections, including on habitat, upon listing. But both have dealt with exemptions in very different ways.

Australia's EPBC Act exemptions

Like the U.S. and Canada, Australia's threatened species law provides for a raft of exemptions to prohibited acts. Under the EPBC Act, the Minister must not issue a permit unless satisfied that:

- The action will significantly contribute to the conservation of the listed species; or
- The impact of the action is incidental to, and not the purpose of, the action, and the action:
 - Will not adversely affect the survival or recovery of the species; and
 - The permit holder will take all reasonable steps to minimise the impact of the action on the species; or
- The action is of particular significance to indigenous tradition and will not adversely affect the survival or recovery of the species; or
- The action is necessary to control pathogens.

One of the greatest threats to Australia's (and Aotearoa's) biodiversity is invasive introduced species. There may be instances where, to address pest species, some collateral damage is inflicted on a listed species. The EPBC does not contain provision to allow non-target mortality of a threatened species under a conservation action.

Experts had to work around that limitation when eradicating pests from Macquarie Island. In that case, a relatively short-term detrimental impact on the threatened northern giant petrel was

predicted by primary and secondary poisoning following the aerial baiting phase of the eradication programme. Annual counts since 2011 have shown that the breeding northern giant petrel population is nearly back to pre-baiting levels. Just five years after the last pest individual was detected on Macquarie Island, BirdLife Australia downlisted eight seabird taxa on the Australian IUCN Red List because the threats justifying a higher threat category were no longer operating.⁵⁸

Spotlight: Overall net gain

A wildlife permit represents harm to a listed species in exchange for a private benefit. In laws aimed at preventing species extinction and enhancing species recovery, a permit is a special privilege that is a cost to society (both present and future) and this cost should be paid by the beneficiary.

Hawaii has embedded the concept of overall net gain in its State law. In Hawaii, all mechanisms that provide some form of exemption to prohibited acts (i.e., habitat conservation plans, safe harbor agreements, incidental takes) must be designed to result in an overall net gain in the recovery of Hawaii's threatened and endangered species. This is coupled with an additional threshold for incidental take permits, which cannot be approved unless the cumulative activities contemplated by the plan are environmentally beneficial.

There is no analogous requirement at the federal level in the U.S. (although President Obama tried, his policy was repealed by President Trump). Instead, the federal system relies on the no jeopardy provision discussed above. However, it does not require additional positive conservation actions and allows for incremental impacts.

England recently mandated biodiversity gain for planning permissions in its new Environment Act 2021. The new Act specifies that biodiversity gain is met if the biodiversity value attributable to the development exceeds the pre-development biodiversity value of the onsite habitat by at least 10 per cent. Figure 7 sets out what the new change adds to existing legislation.

What we are already doing?		What will mandatory biodiversity net gain add?	
Protected by current legislation and planning policy:		Protected by current legislation and planning policy:	
✓ Statutory protected sites ✓ Protected and important species ✓ Irreplaceable habitats		✓ Statutory protected sites ✓ Protected and important species ✓ Irreplaceable habitats	
	Important habitats, such as native woodland, are afforded protection through planning policy only. Sometimes compensation is provided for their loss, depending on case circumstances and local planning policy.		Important habitats, such as native woodland, are afforded protection through planning policy and biodiversity net gain. Losses of important habitats must be compensated with creation or enhancement of the same habitat type.
	Other habitats, such as scrub, may be recognised in local planning policy. They are sometimes, but often not, subject to compensation requirements.		Other habitats, such as scrub, will have their biodiversity value recognised, encouraging the avoidance of impacts and meaning losses must be compensated.

Figure 7. What mandatory biodiversity net gain adds to existing legislation and planning policy⁵⁹

⁵⁸ Garnett et al, 2018, 49-50, 54

⁵⁹ Department for Environment Food and Rural Affairs, 2022

https://consult.defra.gov.uk/defra-net-gain-consultation-team/consultation-on-biodiversity-net-gain-regulations/supporting_documents/Consultation%20on%20Biodiversity%20Net%20Gain%20Regulations%20and%20Implementation_January2022.pdf

Determining overall net gain can be challenging, especially when there is insufficient data on a species (see hoary bat example discussed above). In such cases adaptive management responses, underpinned by a precautionary approach can help overcome issues.

Lessons learnt: Prohibited acts and exemptions

1. Prohibited acts provide a mandatory level of protection for listed species, regardless of other mechanisms (i.e., habitat protection and recovery planning).
2. Some exemptions to prohibited acts are necessary, but these should be limited.
3. The recovery of a listed species should be enhanced by any exemptions issued; there should be an overall net gain to the species.
4. Exemptions to prohibited acts should be accompanied by a carefully framed plan that is scientifically informed, includes qualitative measures and is robustly monitored and enforced.
5. Exemptions need to be considered on a cumulative basis.

Habitat protection

Habitat loss and degradation are the biggest threat to species worldwide. Identifying and protecting critical habitat is essential to a species' survival as a species cannot be conserved in the absence of its habitat. Protecting habitats required for the recovery of a listed species is a major focus of threatened species legislation globally.⁶⁰

Habitat protection which follows a species (such as highly mobile fauna areas in Aotearoa's proposed NPSIB) is different to national reserve or park because actions undertaken within the protected area need to specifically consider the conservation needs of the species. A prime example is 'critical habitats' designated under the U.S., Canadian and Australian systems.

Critical habitat in the U.S.

In the U.S., when a species is classified as endangered or threatened, critical habitat for that species is concurrently designated. Critical habitat does not include the entire geographical area which can be occupied by the listed species, but includes areas that are:

- Within the geographical area occupied by the species at the time it is listed which have physical or biological features essential to the conservation of the species and which may require special management considerations. These features include:
 - Space for individual and population growth and for normal behaviour;
 - Cover or shelter;
 - Food, water, air, light, minerals or other nutritional or physiological requirements;
 - Sites for breeding and rearing offspring, germination, or seed dispersal; and
 - Habitats that are protected from disturbances or are representative of the historical geographical and ecological distributions of the species.
- Outside the geographical area occupied by the species at the time it is listed but which are essential for the conservation of the species. This could account for climate change induced migration and be useful as a supply of sites for compensation or offsetting.

⁶⁰ Palm et al, 2020

In the U.S., critical habitat designations are publicly notified. Final decisions are made based on the best scientific data available, after taking into consideration the probable economic and other impacts of the designation. This has resulted in some critical habitat designations being delayed or stopped if it unduly interferes with economic development or national security (unless that would result in the extinction of the species). Designating critical habitat is particularly contentious for species whose ranges overlay with economically valuable natural resources.⁶¹

A critical habitat designation imposes a duty on all federal agencies to consult with the U.S. Fish and Wildlife Service when authorising, funding or carrying out any activity which could destroy or modify that habitat, i.e., a jeopardy assessment is undertaken. The designation does not affect activities by private landowners if there is no federal nexus. Nor is it linked with the U.S. planning system.

Because of the regulatory impact of a jeopardy assessment (it is difficult to get a jeopardy exemption from the God Squad), there is a significant amount of controversy in the U.S. over where critical habitat is mapped. However, as noted above, the jeopardy assessment has very little impact because federal agencies are granting activities on an incremental basis, rather than examining the cumulative effect of critical habitat destruction or modification.

Outside of critical habitat areas, federal agencies must still consult with the U.S. Fish and Wildlife Service if their actions 'may affect' listed species.

Critical habitat in Canada

Canada similarly requires that critical habitats which are 'necessary for survival or recovery' be identified for listed species. Identification of critical habitats is based on the best available information, including information provided by COSEWIC.

The identification of critical habitats is a crucial part of the recovery process under SARA. Recovery strategies and action plans are tasked with delineating such areas (and action plans must include measures proposed to protect them). Requiring the identification of critical habitat in recovery instruments creates a significant bottleneck in recovery planning. Issues involve a lack of adequate data and political push back, as critical habitat is often at odds with social, economic and political interests.

In the paper '*Fixing the Canadian Species at Risk Act: identifying major issues and recommendations for increasing accountability and efficiency*' it is suggested that critical habitat identification no longer be required for recovery strategies or action plans for species whose habitats are non-threatened. Where there is significant doubt about whether habitat loss or degradation are or could become important threats, critical habitat identification should still be required.⁶²

Once identified, no person can destroy critical habitat if:

- The critical habitat is on federal land, in the exclusive economic zone of Canada or on the continental shelf of Canada;
- The listed species is an aquatic species;
- The listed species is a species of migratory birds protected by the Migratory Birds Convention Act 1994; or

⁶¹ Palm et al, 2020

⁶² Turcotte et al, 2021, 1474–1494

- The critical habitat is on provincial or territorial crown lands and relates to a species that is not an aquatic species or a species of migratory birds protected by the Migratory Birds Convention Act 1994.

Critical habitat designations have very little regulatory impact on private land, restricted to the habitat of migratory birds. There is no analogous jeopardy rule in Canada. Exemptions to the destruction of critical habitat can be issued on the same terms set out above for prohibited acts.

Critical habitat is linked into Canada's planning laws so that projects that are likely to affect a listed species or its critical habitat must be notified to the Minister (albeit there is no requirement for that Minister to do anything). If the project is carried out it must ensure that measures are taken to avoid or lessen effects and to monitor them. The measures must be consistent with any relevant recovery document.

Emergency orders, identifying habitat necessary for the survival or recovery of a species and prohibiting its destruction, are also provided for but have only been exercised twice (in relation to the Greater Sage-Grouse and for the Western Chorus Frog). As noted above, a potential emergency order is also under review for Blanding's turtle.

As of 2015, despite the statutory requirement to identify critical habitat for listed species, only 44 percent of species listed under the ESA, and less than 12 percent of species listed under SARA, had fully identified critical habitat.⁶³

SARA provides for the protection of a species' 'residence' immediately upon listing. While residence is more narrowly defined than critical habitat, protection of these sites provides at least some cover until recovery instruments map the wider habitat requirements of the species.

Residence is defined as follows:

"A dwelling-place, such as a den, nest or other similar area or place, that is occupied or habitually occupied by one or more individuals during all or part of their life cycles, including breeding, rearing, staging, wintering, feeding or hibernating."

Fish have not typically benefitted from protection of residence, thus listed fish species are not afforded any interim protection until (if ever) their critical habitat is designated. Exceptions can be granted as per critical habitats.

Critical habitat in Australia

Unlike the U.S. and Canada, the designation of critical habitat is not mandatory under federal law in Australia and only five critical habitats are registered.⁶⁴ All were registered between 2002 and 2005. Less than 1 percent of listed species had critical habitat designated and protected as of 2017.⁶⁵

Victoria's recently amended Flora and Fauna Guarantee Act 2019 requires identification of critical habitat. While there are no regulatory implications upon determination, the Secretary is obliged to try and manage the land co-operatively with landowners. Critical habitat can, however, be made subject to a habitat conservation order. Such orders, made by the Minister, can prohibit damage to critical habitat or require remediation of previous damage. Habitat conservation orders can be

⁶³ Palm et al, 2020

⁶⁴ <https://www.environment.gov.au/cgi-bin/sprat/public/publicregisterofcriticalhabitat.pl>

⁶⁵ Martin et al, 2017, 10, 308–316

appealed, and persons affected by them are entitled to compensation for financial loss if the order affects an existing use right, or a pre-existing approval. It is an offence to contravene a habitat conservation order, with potential jail time of two years.

Queensland's Nature Conservation Act 2002 enables the Minister to issue an interim conservation order if critical habitat (or threatened wildlife) is subject to a threatening process that is likely to have significant detrimental effect on the area (or species). An interim conservation order is limited to 60 days operation but it can provide for the prohibition or control of a threat during that time, including on land that is not critical habitat but which requires threat abatement measures for the conservation, protection or management of the critical habitat. If a critical habitat is the subject of an interim conservation order the landholder is entitled to be paid reasonable compensation by the State.

Australia also has Indigenous Protection Areas (IPA) which protect biodiversity, as well as providing economic, social and cultural benefits to indigenous communities. IPAs contribute almost one half of the National Reserve System, principally in arid lands and northern Australia.⁶⁶ Early IPA management plans were criticised for reflecting a western approach to conservation, valuing biodiversity and heritage over the objectives of the traditional owners. In response, guidelines were created to ensure IPA planning reflects the cultural context.⁶⁷

Natura 2000 protected areas network

Under the European Union's Birds and Habitat Directives, member states are required to designate ecologically special areas. These areas make up the Natura 2000 protected areas network (Figure 8) which comprises one fifth of the European Union's land area and around 10 percent of its marine area.⁶⁸

Annex III of the Habitats Directive sets out the criteria for selecting sites eligible for designation. Of note, the Directive gives specific direction for identifying sites that relate to species ranging over wide areas. The focus is on areas essential to life and reproduction (akin to critical habitat):

- For animal species ranging over wide areas - places within the natural range of a species which present physical or biological factors essential to their life and reproduction.
- For aquatic species ranging over wide areas - where there is a clearly identifiable area representing the physical or biological factors essential to their life and reproduction.

Sites are meant to be identified based on scientific information only, but some interviewees indicated that vested interest interference occurs.

The Habitats Directive requires that Member States take appropriate conservation measures to maintain and restore the habitats and species for which the site has been designated to a favourable conservation status.

The concept of favourable conservation status is significant for the Directives because it is the benchmark for when conservation measures are required.⁶⁹ While there has been extensive review of the definition of recovery under Acts like the ESA, the concept of favourable conservation status

⁶⁶ Marsh, 2020, *Submission to the EPBC Act review*, Threatened Species Scientific Committee, March 2020

⁶⁷ Davies et al, 2013

⁶⁸ <https://www.eea.europa.eu/themes/biodiversity/state-of-nature-in-the-eu/natura-2000-networks-contribution-to>

⁶⁹ Although the Birds Directive uses slightly different language, the European Commission has advised it applies equally to that Directive

has received far less scrutiny. The European Commission has released guidance on the topic, but it remains a pressure point for implementation of the Directives.

Under the Birds Directive (which applies to birds, their eggs, nests and habitats), the species' population must be maintained at a level that corresponds to ecological, scientific and cultural requirements, while taking account of economic and recreational requirements. Measures to achieve this include creating protected areas to preserve, maintain or re-establish a sufficient diversity and area of habitat for birds based on:

- Species in danger of extinction
- Species vulnerable to specific changes in their habitat
- Species considered rare because of small populations or restricted local distribution
- Other species requiring particular attention for reasons of the specific nature of their habitat

Under both Directives, Member States must avoid damaging activities that could significantly disturb species or deteriorate the habitats of the protected species or habitat types.

Further, any plan or project likely to have a significant effect on the site (including cumulatively) can only be approved if it will not adversely affect the integrity of the site and, if appropriate, after having obtained public opinion. The threshold of 'significant' is an issue for the Directives, with Member States interpreting it differently.

Notwithstanding a negative assessment, a plan or project can still be carried out if there are overriding public interest reasons. This bar is higher for sites that host priority natural habitats and species. In those cases, a plan or project with a negative assessment can only proceed on the basis of human health, public safety, beneficial consequences of primary importance for the environment, or overriding public interest (in the Commissioner's opinion).

Measuring the effectiveness of the Natura 2000 network is limited because of the lack of appropriate monitoring. The European Union reports that land cover changes are less within Natura 2000 sites than outside, but habitats are still being lost, particularly grasslands. There remains an ongoing need to strengthen conservation measures by Member States. That includes development of management plans for Natura 2000 sites. In the absence of such plans, the site could be considered a 'paper park'.

The Court of Justice of the European Union has, in a long series of judgements, made it clear that the Directives must be strictly followed. Including by adopting a strong precautionary approach, approving projects on protected sites only if there is no scientific doubt that they can proceed without adverse effects. These judgements have been a key tool for nature conservation. However, implementation and enforcement of the cases remains an issue within Member States.⁷⁰

⁷⁰ See Implementation of rulings for nature conservation at <https://eeb.org/wp-content/uploads/2020/09/Implementation-of-rulings-for-nature-conservation2.pdf>

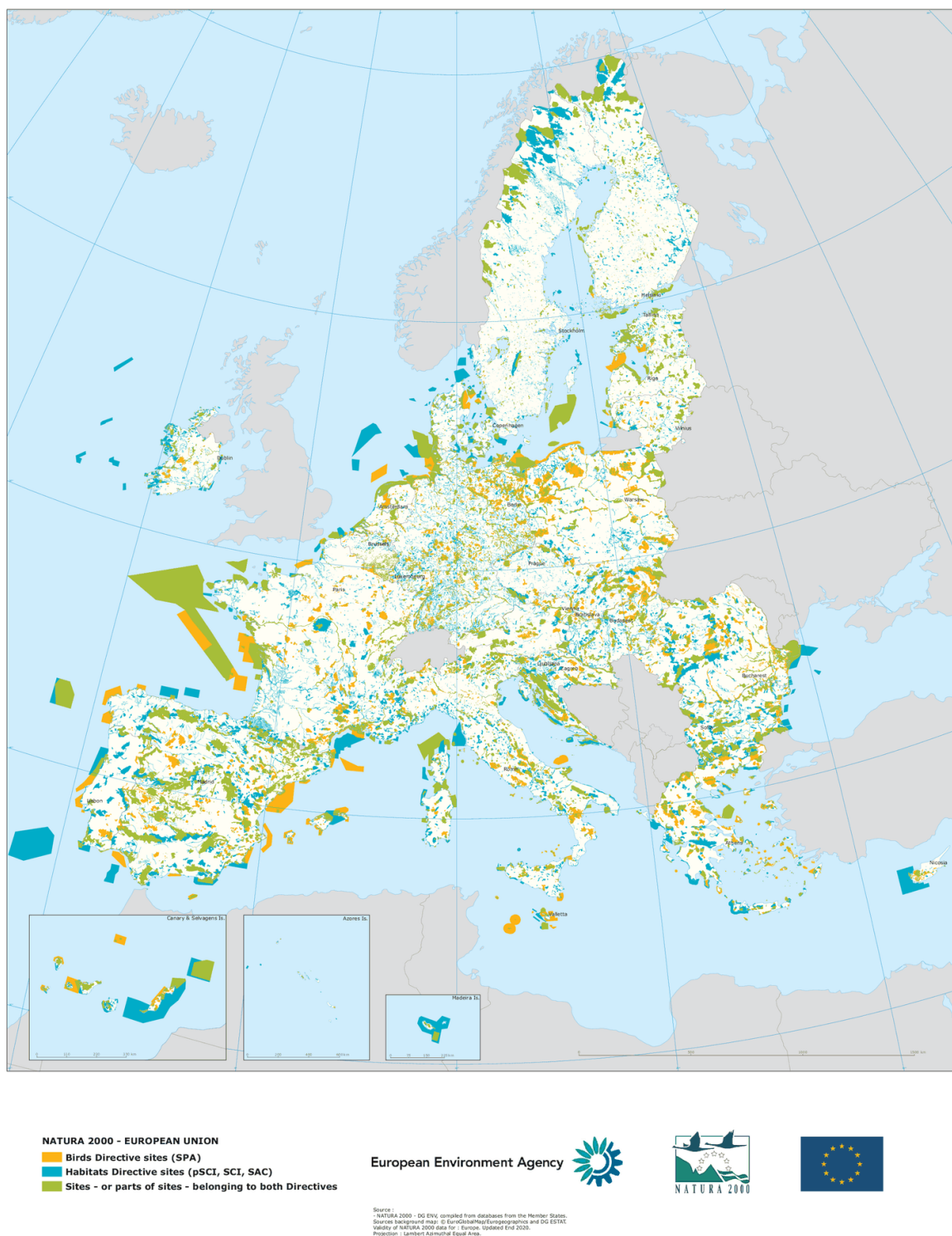


Figure 8. Natura 2000 Birds and Habitat Directive sites

Spotlight: Finland's protections on private land

Under Finland's Nature Conservation Act 1996, any species at imminent risk of extinction can be strictly protected such that the deterioration or destruction of habitat important for the survival of the species is prohibited. In consultation with landowners, the Centre for Economic Development, Transport and the Environment sets the boundaries of a site hosting a species under strict protection. Landowners can appeal the Centre's decision. Derogations from the protective provisions can also be applied for provided that the 'conservation status of the affected species remains favourable'.

Further, the Act provides for 'nature reserves' to be identified within nature conservation programmes. These reserves can be established by the Government on private land, even without the landowner's consent. Actions which jeopardise the conservation objectives of a nature reserve are prohibited. Protections instituted in such reserves must be consistent with the nature conservation programme. If a property owner affected by a nature reserve incurs 'significant inconvenience' they are entitled to full compensation from the State, but only after an exemption has first been applied for.

Listing ecological communities in Australia

In Australia, a group of native plants, animals and other organisms can be identified and protected as a Critically Endangered (extremely high risk of extinction in the wild in the immediate future), Endangered (very high risk of extinction in the wild in the near future) or Vulnerable (high risk of extinction in the wild in the medium term) ecological community under the EPBC Act. Listing a threatened ecological community follows the same route as listing threatened species; the TSSC makes a recommendation to the Minister (based on IUCN criteria) who decides whether to list the area.

A listed ecological community is a matter of national environmental significance, thus actions likely to have significant impact require Ministerial approval. Such activities may include:

- Changes to natural drainage regimes, such as the diversion of water, affecting the ecological community.
- Clearing of the ecological community, dumping of spoil, construction of structures fragmenting the community or impeding natural water balances (e.g., causeways, raised fence lines).
- Clearing of native vegetation adjacent to the listed community or in the immediate upstream catchment such that drainage regimes supporting the ecological community are affected.
- Land reclamation and/or dredging in areas where a listed ecological community occurs, or in the vicinity of the ecological community such that parameters critical to the survival of the ecological community are affected.
- Significant and adverse changes in management regimes affecting the community, including too frequent or too infrequent fire.
- New weed management regimes that pose significant risk to the listed community (e.g., aerial spraying).
- Allowing new access for domestic stock and other grazing animals (e.g., where there has previously been no access) or significant intensification in the numbers of animals with access to the ecological community.

requires a more formal process for generalising knowledge for recovery planning,⁸¹ and knowledge of the location of future climatically suitable habitat.⁸²

Self-assessment by proponent proposing to take action

Between 2000 and 2017, 7.7 million hectares of potential habitat for terrestrial threatened species has been lost in Australia and 370,000 hectares of threatened ecological communities have been lost. Of this loss, 93 percent was not referred to the federal Government for assessment, meaning the loss was illegal. The loss included 2.3 percent of the koala's habitat. Listed species with wide ranging, common, but declining populations are at significant risk from illegal and unreported landowner actions.

Ecosystem habitat protection

Since 1975,⁸³ the scientific community has called for an ecosystem approach to conservation which protects all species and ecological processes across entire landscapes.⁸⁴ Considering the health and sustainability of entire ecosystems is a more comprehensive approach, rather than piecemeal protection of individual species.⁸⁵ It is also preventative, rather than only responding when species are near the point of extinction.⁸⁶ An ecosystem approach also avoids unintended and perverse outcomes, like conflict or competition between protected species,⁸⁷ or using harmful chemicals to narrowly focus on weed eradication without considering the wider ecological effects.⁸⁸

Some are wary that such a broad approach will overlook threatened species. However, both threatened and common species benefit under an ecosystem scale approach.⁸⁹ Although it may initially be more expensive than single-species protection, this is offset in the long-term. Tackling extinction on a species-by-species basis can be ineffective, and preparing multiple recovery programs is time consuming and resource intensive.⁹⁰

Ecosystem scale protection is better suited to complex problems like water quality or climate change.⁹¹ Rather than focusing on case-by-case management, an ecosystem approach takes into account the potential for cumulative or synergistic effects of multiple activities.⁹² In the face of the climate crisis, ecosystem-scale translocation, rather than translocating single species, increases resilience through the conservation of complete, functioning communities.⁹³

Lessons learnt: Habitat protection

1. Habitat protection is critical to achieving threatened species conservation.
2. Comprehensive and accurate definition of critical habitat can be difficult, particularly for wide ranging species.

⁸¹ Rumpff, 2021, Conservation actions for Threatened Ecological Communities', Threatened Species Recovery Hub <https://www.nespthreatenedspecies.edu.au/projects/conservation-actions-for-threatened-ecological-communities>

⁸² Harris et al, 2015

⁸³ Cryer et al, 2015, 57-70

⁸⁴ Bader et al, 1992, 137-147

⁸⁵ Peggy Fiedler et al, 1993, 7-11

⁸⁶ Welner, 1995, 319-361

⁸⁷ Roemer et al, 2003, 1251-1260

⁸⁸ Williams et al, 2001, 425-444

⁸⁹ Fiedler et al, 1993, 7-11

⁹⁰ Bader et al, 1992, 137-147

⁹¹ Belfiore et al, 2006

⁹² Taljaard et al, 2006, 535-542

⁹³ Boyer et al, 2016, 1-11

3. Critical habitat identification is routinely influenced by economic, social, and cultural considerations.
4. Critical habitat designations have been applied internationally to trigger effects assessment of activities on Crown land or involving Government agencies.
5. Critical habitat designations have very little regulatory impact on private landowners except in the USA.
6. Critical habitat designation has not been very effective at preventing habitat destruction or modification, either because it has not been designated, or because effects have been assessed on a case-by-case basis and cumulative effects ignored.
7. Designating critical habitat upon listing provides immediate protections.
8. Critical habitat identification needs to factor in climate change induced movement.
9. When critical habitat identification has been required by recovery instruments, it has significantly delayed recovery efforts.
10. A network of protected sites and areas has the potential to provide landscape scale habitat protection.
11. Protection of habitat is likely more effective at the ecosystem scale.

Recovery planning

Recovery planning is a significant component of threatened species laws in the jurisdictions reviewed. Recovery instruments are meant to pull together all the conservation actions and habitat requirements for the species.

The value of recovery plans, strategies or actions in rehabilitating a species' population is closely tied to the extent to which these instruments are informed and dictated by science, which sets clear targets for recovery and trigger points for conservation intervention. Recovery documents then need to be implemented properly and supported by sufficient funding and resourcing.

*"When recovery plans are deemed necessary, it is crucial that they are funded and implemented, with clear accountability for actions outlined and enforced, and thorough evaluation of management effectiveness. Evaluations of recovery plans and associated data should be compiled in a central depository for public access."*⁹⁴

Further, to be effective, recovery programmes need to address threats to the subject species. Recovery actions that fail to stop threatening processes may be well-intentioned but are ultimately futile (see discussion of Australia's key threatening processes below).

Recovery plans appear to have worked best for obscure species with small ranges whose threats are easily identified and able to be addressed. They are less successful for wide ranging species facing multiple threats.

Mandatory vs voluntary

Under the ESA, federal agencies must develop recovery plans for all listed species (unless such plans will not promote the conservation of the species). A U.S. recovery plan is intended to set out recommendations that are based on resolving the threats to the species and ensuring self-sustaining populations in the wild. The plan must incorporate:

- A description of such site-specific management actions as may be necessary to achieve the plan's goals for the conservation and survival of the species;

⁹⁴ Scheele et al, 2018, 668-675.

- Objective, measurable criteria which, when met, would result in a determination that the species be removed from the list; and
- Estimates of the time required and cost to carry out those measures needed to achieve the plan's goals and to achieve intermediate steps toward that goal.

The Act requires that the Secretary report every two years on the status of efforts to develop and implement recovery plans. There is no statutory check to ensure that recovery plans are not deficient.

While recovery planning is compulsory under the ESA, there is no timeframe for final completion. As of October 2020, there were more than 600 draft and final recovery plans for 1,666 species listed in the U.S. One fourth of those listed species lacked final recovery plans, and more than half of recovery plans were more than 20 years old. Funding was the main constraint on their completion.⁹⁵ Delays in recovery plans is a recognised barrier in endangered species conservation.⁹⁶ It can lead to increased costs, uncertainty for landowners and industry, and lost opportunity for species recovery. For this reason, some argue that conservation actions should be triggered by a species' assessment and listing, and not be left to recovery instruments.

Further, there is no regulatory imperative to implement recovery plans in the U.S. Undertaking recovery actions is discretionary, even for federal agencies. The lack of deadline for finalising recovery plans and their weak regulatory impact are significant issues contributing to species-level biodiversity loss in the U.S.

Canada's SARA similarly requires recovery strategies but goes further and requires action plans to implement the strategies. Canada's recovery strategies must include information on threats to the listed species and its habitat, identification of critical habitat, and a statement of the population and distribution objectives that will assist the recovery and survival of the species. Reporting on implementation of such strategies is undertaken five yearly. Action plans set out the actions required to implement recovery strategies. They can consider the socio-economic costs of proposed actions.

Canada's two-step recovery planning process has the advantage of clearly separating scientific recommendations from management actions. However, in practice the preparation of these two documents causes serious delays. As of 2017, the average time for completion of a recovery strategy exceeded six years.⁹⁷ As noted above, not all of Canada's listed species have recovery strategies or action plans.

In contrast to the U.S. and Canadian systems, recovery planning in Australia is not mandatory (it was until 2006). In 2019, of the 449 listed fauna species in Australia only 45 percent had recovery plans. Permits, agreements and declarations must not be inconsistent with a recovery plan for a species, however, there remains no obligation to implement them. Further, without any monitoring of recovery plans, there is no transparency about their effectiveness.

In the absence of a recovery plan, a threatened species is reliant on non-binding conservation advice; the Minister need only have regard to conservation advice before making decisions under the EPCB Act. Conservation advices identify actions required for conservation and recovery of the threatened species or ecological community and inform the Government's regulatory decision-making and investment in threatened species and ecosystems. In 2019, only 75 percent of listed

⁹⁵ Congressional Research Service, 2021, 25

⁹⁶ Kraus et al, 2021, 1088-1127

⁹⁷ Westwood et al, 2019

fauna species had conservation advice. Three species were not covered by either a recovery plan or conservation advice.

Notwithstanding those statistics, in the book *'Recovering Australian's Threatened Species: A Book of Hope'*, recovery plans were often listed as factors in the successful recovery of listed species. Such plans were however extensively supported with long-term funding and a dedicated recovery team. Recovery teams are typically a mix of government agencies, non-government organisations, scientists, industry, and the community.

Australia has prepared a draft national reporting framework for recovery teams which is currently being trialed. A national network of recovery teams, to connect recovery planning practitioners, is also under consultation. Without proper implementation a recovery plan is meaningless.

Recovery is informed by science

Recovery planning was introduced into the ESA in 1988 as a means of organising conservation efforts to prevent extinctions and recover populations. They were intended to be a significant component of the Act's implementation. Although recovery plans have evolved significantly over the years, practitioners recognise that significant challenges remain with the recovery planning process.⁹⁸ A major challenge is the lack of definition, statutory criteria or clarity on what 'recovered' actually means.

In 2020, a U.S. study found that in 130 cases, the U.S. Fish and Wildlife Service could consider a listed species to be recovered even though it had fewer populations or individuals than existed at the time of listing.⁹⁹ Fewer than half of the plans for those 130 species had scientific support for a 'recovered' decline.

As delisting a species due to recovery is a measure of success for the ESA, a lower recovery threshold is helpful from a political standpoint because success can be claimed despite actual failure. Consequently, federal agencies have made the most of the Act's discretion when defining recovery by lowering the bar for success.

Statutory guidance and accountability is required to ensure consistency and clarity of recovery criteria and thresholds. In 2019, the U.S. Department of Interior stated that it would require recovery plans to include quantitative criteria that specify when a species has recovered, which would aid in transparency in delisting decisions.¹⁰⁰ Just as science (and indigenous knowledge) should dictate listings, so too should it dictate delistings.

Recovery planning in Canada is unusual in that COSEWIC does not have a role in approving recovery plans, whereas analogous scientific bodies in the U.S. and Australia do. This means that while COSEWIC applies a consistent and arm's length framework for extinction risk assessments by independent experts, no counterpart exists at the recovery strategy or action planning stage. It is unclear, therefore, whether the Government's recovery planning is based on the best biological information available.¹⁰¹ There have been instances where the Courts have become the arbiter in determining whether recovery strategies are in accordance with SARA's purpose, following intervention by non-governmental organisations who flagged problems with such strategies. Including the threat assessing body in recovery directions streamlines the process of species protection and reduces bias and political influence in recovery decisions.

⁹⁸ Malcom et al, 2018, 1-9

⁹⁹ Recovery plans need better science to support decisions that allow species to decline in abundance but be recovered, 2020

¹⁰⁰ Congressional Research Service, 2021, 25-26

¹⁰¹ Turcotte et al, 2021, 1474–1494

Indigenous and community led conservation

In Canada, recovery strategies must be prepared in co-operation with every Aboriginal organisation that the competent Minister considers will be directly affected by the recovery strategy. The direction is to prepare plans and co-operate “to the extent possible”.

Despite this statutory recognition, there has generally been an inadequate co-operation between federal agencies and indigenous peoples regarding species at-risk management.¹⁰² For example, a 2019 study reported that only 50 percent of recovery planning documents demonstrated indigenous involvement, despite the legal requirement to collaborate to the extent possible with aboriginal organisations.¹⁰³ The lack of clear guidelines on the engagement process has led to a deficiency in authentic participation.¹⁰⁴

One suggested way to increase partnerships is to initiate conservation agreements. Conservation agreements could recognise the authority of indigenous peoples over listed species, species at-risk, or habitats and provide financial support and compensation for conservation stewardship activities.¹⁰⁵

Monitoring and responses

Monitoring should be considered a pivotal component of a species’ recovery response.¹⁰⁶ In Australia, there is no integrated monitoring for biodiversity and few threatened species are adequately monitored.¹⁰⁷ In 2018, between 21 - 46 percent of threatened vertebrate species received no monitoring at all. Where monitoring did occur, it was found to be poorly designed, with low power to detect changes in population abundance or to quantify the efficacy of management actions. The EPBC Act has no requirement to implement or report on monitoring. This is an area of proposed reform by the TSSC, which relies on good data to make informed listing and recovery decisions.

Australia is now introducing a national standardised monitoring protocol and data management standard to promote best practice recovery actions. According to Australia’s Threatened Species Strategy Action Plan, the monitoring protocols will be used to capture consistent baseline and ongoing time-series data for all priority species and places compatible with national data standards. They will be progressively included in updates to conservation planning documents, including recovery plans.

Australia’s Threatened Species Recovery Hub has developed a world first threatened species index which combines data from monitoring programmes across the country to track trends in species populations. The index has identified trends like, in the absence of active conservation, populations of threatened mammals have decreased by on average 60 percent over the last 20 years. But that actions, such as feral animal control, feral predator exclusion fencing, and ecological fire management have resulted in an increase in populations.¹⁰⁸ It is hoped that the index will assist with prioritising threat abatement and recovery efforts.

¹⁰² Turcotte et al 2021, 1474–1494

¹⁰³ Hill et al, 2019, 220-226

¹⁰⁴ Turcotte et al, 2021, 1474–1494

¹⁰⁵ Turcotte et al, 2021, 1474–1494.

¹⁰⁶ Woinarski, et al, 2016, 13-23.

¹⁰⁷ Woinarski et al, 2016, 13-23.

¹⁰⁸ Threatened Species Recovery Hub, 2020, ‘Major declines in threatened mammal populations over last 20 years, but news not all bad’, <https://www.nespthreatenedspecies.edu.au/news-and-media/media-releases/major-declines-in-threatened-mammal-populations-over-last-20-years-but-news-not-all-bad>

Informed by monitoring, recovery instruments need qualitative measures which trigger adaptive management, emergency response or remedial interventions required. There is no point monitoring a species to extinction, as has occurred in Australia. Monitoring programmes need triggers for management interventions.

Factoring in climate change

Threatened and endangered species are likely to be disproportionately affected by climate change because they are often habitat specialists and rare.¹⁰⁹ Yet recovery plans and management responses often do not account for climate change. In the U.S., agencies consider climate change as a threat to 64 percent of species but plan management actions for only 18 percent.¹¹⁰

Just as there are tools to assess species that may become at-risk due to climate change, there are also tools that can assess its impact on listed species and their habitats. This information should be factored into recovery plans with resultant actions, including preparing a species for assisted migration or 'climate banking', where future critical habitat is secured.

A significant turning point in the recovery of the red-tailed black-cockatoo in Australia was shifting focus from the bird's existing population and habitat to a future habitat regeneration model. The new outlook inspired landscape restoration projects which have protected and/or re-established hundreds of hectares of feeding habitat for the red-tailed black-cockatoo.¹¹¹

Catchment based recovery planning

With the multitude of species currently listed, and the anticipated flood of newly listed species as climate change bites, it seems an impossible task to continue doing recovery planning on a species-by-species basis. Countries already cannot clear the backlog for currently listed species in terms of updating old plans and developing new ones. The answer seems clear; a more strategic approach is necessary.

Internationally, commentators have called for a more cost-effective ecosystem-based approach where sympatric species with similar biological requirements and threats can be addressed by the same actions.¹¹²

Although not typically statutorily mandated, regional, catchment or watershed recovery planning is already underway in some jurisdictions. Maui in Hawaii is currently preparing an island-based plan which aims to identify the habitats required for recovery of listed and non-listed species and prioritise threat abatement.

Catchment-based recovery planning has been implemented to good effect in Australia. In the Mary River catchment in south-east Queensland recovery planning for one threatened species, the Mary River cod, spurred the listing of several other species within the catchment. Recovery programmes for individual species' ran in parallel to each other until a dam proposal threatened the river and a multi-species regional recovery plan was developed. Almost 60 species in the catchment are now listed under the EPBC Act and management actions are undertaken in a more holistic way.¹¹³

¹⁰⁹ Kraus et al, 2021, 1088-1127

¹¹⁰ Kraus et al, 2021, 1088-1127

¹¹¹ Garnett et al, 2018, 150

¹¹² Noss et al, 2021, 1715-1724

¹¹³ Garnett et al, 2018, chapter 5, 71-82

Many have called for more strategically mandated recovery planning. Experts in British Columbia have recommended that the new provincial species at-risk law has an independent scientific committee which groups species for recovery action and coordinates the recovery planning process. Providing for multi-species recovery plans and actions to be strategically developed.

In Australia, the TSSC recommended regional recovery plans in its submission on the EBPC Act. England's new Environment Act 2021 requires nature recovery strategies, which are to include:

- A statement of biodiversity priorities for the area (description of opportunities for recovery, recovery priorities and measures to implement those priorities).
- A local habitat map for the area, identifying Crown owned nature areas, areas that are/could become important for biodiversity and areas where the recovery or enhancement could make a contribution to other environmental benefits.

The UK's Department for Environment, Food and Rural Affairs is currently investigating what future role nature recovery strategies should play. For example, a role in the land use planning system, and in supporting England's net gain requirement.

Although regional or catchment-based recovery planning seems the way of the future, many interviewees cautioned that such high-level planning risks overlooking individual species. It seems that both approaches are required, habitat level and species level recovery, with ecosystem recovery sitting above that.

European Commission's Nature Restoration Law

In 2022, the European Commission proposed a Nature Restoration Law to give effect to the EU Biodiversity Strategy 2030, which calls for targets to restore degraded ecosystems and species populations. The proposal contains the following specific targets:

- Targets based on existing legislation (for wetlands, forests, grasslands, river and lakes, heath & scrub, rocky habitats and dunes) - improving and re-establishing biodiverse habitats on a large scale, and bringing back species populations by improving and enlarging their habitats.
- Pollinating insects - reversing the decline of pollinator populations by 2030, and achieving an increasing trend for pollinator populations, with a methodology for regular monitoring of pollinators.
- Forest ecosystems - achieving an increasing trend for standing and lying deadwood, uneven aged forests, forest connectivity, abundance of common forest birds and stock of organic carbon.
- Urban ecosystems - no net loss of green urban space by 2030, and an increase in the total area covered by green urban space by 2040 and 2050.
- Agricultural ecosystems - increasing grassland butterflies and farmland birds, the stock of organic carbon in cropland mineral soils, and the share of agricultural land with high-diversity landscape features; restoring drained peatlands under agricultural use.
- Marine ecosystems - restoring marine habitats such as seagrass beds or sediment bottoms that deliver significant benefits, including for climate change mitigation, and restoring the habitats of iconic marine species such as dolphins and porpoises, sharks and seabirds.
- River connectivity - identifying and removing barriers that prevent the connectivity of surface waters, so that at least 25,000 km of rivers are restored to a free-flowing state by 2030.

Under the law EU countries will have to submit National Restoration Plans to the Commission, demonstrating how they will deliver on the targets. Monitoring and regular reporting will also be part of the package to ensure effective implementation.

Lessons learnt: Recovery planning

1. It is important that recovery planning is mandatory, timeframe specific, has regulatory impact and accountability.
2. Required recovery actions should be informed by science, but management responses may factor in other considerations.
3. Robust recovery instruments include monitoring requirements, and objective triggers for management interventions.
4. Recovery needs to focus on threat reduction.
5. Climate change impacts should be factored into recovery instruments.
6. Wherever possible, catchment-based recovery planning should be implemented, alongside species specific recovery planning.

Addressing threats

Reducing or eliminating threats to threatened species is key to achieving species protection and recovery. As noted above, recovery actions that do not stop threatening processes are well-intentioned but futile. As Andrew Cox, CEO of Australia's Invasive Species Council, put it "[w]e're not looking deeply at what's causing declines and tackling the declines at the source".¹¹⁴ Addressing threats is not just important for recovering threatened species. It is also important to prevent common species becoming threatened.

Australia appears to be the only country in the world with a formal national threat abatement system. Whereas other jurisdictions focus on threatened species and address threats through recovery actions, Australia has a specific process for listing threats, separate to its species recovery mechanisms. This could be a powerful strategic tool for saving threatened species, but the system has been poorly applied.¹¹⁵

Under the EPBC Act, the Minister must establish a list of 'key threatening processes' which are processes that threaten, or may threaten the survival, abundance or evolutionary development of native species or ecological communities by either:

- Causing a native species or ecological community to become eligible for listing (or being listed in a category with a higher degree of endangerment); or
- Adversely affecting two or more listed species or ecological communities.

Listing a key threatening process follows a similar process to listing a species. Any person can nominate a key threatening process for listing under the EPBC Act. Nominations are considered by the TSSC, which makes a recommendation to the Minister who decides whether to list the threatening process.

Listing a key threatening process can help to manage land or practices to improve opportunities for biodiversity, however, a key threatening process has no statutory obligations attached to it. For

¹¹⁴ Cox L, 2020, 'Australian government stops listing major threats to species under environment laws, 7 May 2020, <https://www.theguardian.com/environment/2020/may/08/australian-government-stops-listing-major-threats-to-species-under-environment-laws>

¹¹⁵ Invasive Species Council et al, 2020

example, it is not considered a matter of national significance under the EPBC Act, which means that the Minister is not required to consider them when making decisions about projects. This constraint means that key threatening processes are not prioritised. They are only effectual once a threat abatement plan is adopted for the threatening process.

Once a key threatening process is listed a threat abatement plan can be promulgated to abate the threatening process. Such plans establish a national framework to guide and coordinate Australia's response to the threatening process. Government agencies must implement threat abatement plans, and the Minister must not grant or issue anything under the EPBC Act that is inconsistent with them (i.e., they have the same status as recovery plans). Outside of Government implementation, the plans are actioned on a voluntary basis.

There are 22 key threatening processes currently listed under the EPBC Act, including predation by feral cats, land clearance and incidental catch of seabirds.¹¹⁶ With the exception of one, all key threatening processes were listed between 2000 and 2014. In 2022, 'fire regimes that cause declines in biodiversity' was listed as a key threatening processes, 14 years after it was first added to the assessment list in 2008. It seems it was only added in response to the 2019/2020 bushfires.¹¹⁷

In 2001, the EPBC Act listed the 'loss of climatic habitat caused by anthropogenic emissions of greenhouse gases' as a key threatening process but no associated threat abatement plan was ever formulated. A plan was not considered to be the most 'feasible, effective and efficient way to abate the process' (the test for whether an abatement plan is required) as most emissions occurred outside of Australia. Of the processes listed, only 13 have associated threat abatement plans. The Longline Fishing Threat Abatement Plan is lauded as the most successful (see below *Spotlight: Seabird bycatch in longline fishing*).

Key threatening processes and threat abatement plans have been underutilised by the EPBC Act and, until recently, the Government stopped listing threats under the Act.¹¹⁸ A 2019 Government briefing document stated key threatening processes had 'limited regulatory influence' and, therefore, the Department did not have capacity to resource assessments of them.¹¹⁹ Key threatening processes and threat abatement plans suffer the same criticisms associated with listing species and recovery planning under the EPBC Act. Too many delays, too many opportunities for vested interests to influence decision-making, poor implementation and no accountability.

A major impediment to effective threat abatement in Australia is lack of funding. It is estimated that about AU\$1.1 billion a year would be needed to manage high-priority terrestrial threats in Australia. Additional funding would be needed for abating threats in marine habitats and on islands, as well as for species-specific recovery actions. That figure does not include funds required for threat abatement research.

A recent review of the EPBC Act identified the threat abatement system as a key priority for reform. The review noted the following deficiencies with the existing regime:

- Threat abatement plans are not required, and few plans are up to date.
- Cumulative impacts on the environment are not systematically considered.
- Mechanisms to quickly respond to acute threats are lacking.

¹¹⁶Department of Climate Change, Energy, the Environment and Water, 'Listed Key Threatening Process', <https://www.environment.gov.au/cgi-bin/sprat/public/publicgetkeythreats.pl>

¹¹⁷ See Cox, 2020

¹¹⁸ Kraus et al, 2021, 1088-1127

¹¹⁹See Cox, 2020

- Opportunities for coordinated national actions to address key environmental challenges are *ad hoc*, rather than a key national priority.

A 2022 report called '*Averting extinctions, the case for strengthening Australia's threat abatement system*' set out seven ways to improve the system:¹²⁰

- Comprehensively identify and list threats to nature through an independent scientific process and regularly review the list to ensure it remains up to date.
- List threats in a hierarchical scheme of key threatening processes and environmental threats of national significance.
- Establish an additional threat category - emerging threatening processes.
- Design a fit-for-purpose national abatement response for all listed threats, including national and regional plans, and policy and regulatory responses.
- List key threatening processes as matters of national environmental significance.
- Establish an implementation taskforce for each threat response.
- Systematically monitor and report on threat abatement progress.

If properly funded and implemented, threat abatement plans represent a cost-effective mechanism to address threats to Australia's biodiversity.¹²¹ Australia's Threatened Species Strategy Action Plan 2021 - 2026 recognises that, and focusses on threat management where Commonwealth leadership can improve landscape-scale outcomes.

Spotlight: Seabird bycatch in longline fishing

In the late 1980s it became clear that longline fishing was killing thousands (up to 10,000) of threatened seabirds each year in Australia. In 1995, seabird bycatch in longline fishing was listed as a key threatening process. An associated Longline Fishing Threat Abatement Plan was also developed. As a result of these processes, bycatch has been reduced to fewer than 50 birds per year.¹²²

Listing seabird bycatch in longline fishing and the Longline Fishing Threat Abatement Plan are often cited as Australia's most successful threat abatement case. Key reasons for its success are sufficient funding, that there was a technical solution to mitigating the threat, and high stakeholder involvement.

The Longline Fishing Threat Abatement Plan has undergone three iterations since its inception in 1998 and continues to drive conservation improvements. Importantly, the plan was supported by a ministerial advisory body which met annually to review progress in mitigating bycatch in Australian fisheries. The 'TAP team' was instrumental in achieving continued progress.

Lessons learnt: Addressing threats

1. A species' recovery is heavily reliant on reducing threats to its survival.
2. Threat abatement planning can be a cost-effective way of achieving positive landscape-scale conservation outcomes.
3. Threat abatement is most effective when the threat is well defined, there are technical solutions to reducing the threat, abatement actions are well funded and stakeholder engagement is high.

¹²⁰ Invasive Species Council et al, 2020

¹²¹ The Ecological Society's submission to the Environment and Communications References Committee on Australia's faunal extinction crisis

¹²² Garnett et al, 2018, chapter 4, 58-68

4. Climate change is a serious threat to species at-risk that should be addressed at the national scale.

Incentives

*“Conservation will ultimately boil down to rewarding the private landowner who conserves the public interest”.*¹²³

Private land is a significant and important refuge for many listed species. For example, two thirds of all listed species in the U.S. are found on private land. As the number of species and habitats subject to protection and recovery increases, so too do perverse incentives to clear habitat or harm listed species to avoid compliance with regulations. Government funded stewardship programmes, which are more directive, flexible and incentives based, can counteract that effect. Incentives was identified by the IPBES as one of the five main interventions required to generate transformational change in biodiversity loss.

The ESA has a long history of negotiating and implementing conservation on private lands. In addition to supporting regulatory compliance, two main initiatives, Safe Harbor Agreements and CCAA, foster cooperation between stakeholders so that despite a focus on federal regulation, the U.S. has overcome many stewardship challenges related to listed species on private lands.¹²⁴ Both Safe Harbor Agreements and CCAA provide landowners with certainty that their voluntary actions to improve habitat or increase species numbers will not result in additional regulation. The Safe Harbor Programme has been particularly successful in restoring Red-cockaded Woodpecker populations in southeastern U.S. (see *Spotlight: Recovery of the Red-cockaded Woodpecker*).¹²⁵

*Spotlight: Recovery of Red-cockaded Woodpecker*¹²⁶

Red-cockaded Woodpecker live exclusively in pine forests in southeastern U.S. Population estimates are based on the number of breeding grounds, or active clusters, rather than individual birds. It is estimated that at the time of European settlement there were between 1 - 1.6 million active clusters. At the time of listing as an Endangered species in 1970 the population was down to 3,000 - 4,000 clusters. The primary threat to the species is the lack of suitable habitat. The birds require mature pine forests which comprise trees old enough so that they can excavate cavities to live in.

Conservation efforts began in 1979 with a recovery plan but little progress was made and in 1990 the population had declined to just 1,981 active clusters. Fearing regulation under the ESA, many private landowners actively managed their forests to avoid cutting down mature trees. In 1992, realising that private lands would need to play a significant role in the survival of the species, a private lands conservation strategy was developed. The strategy arose when the Defense Department, an owner of a large area of old forest, was unable to conduct training exercises in the Woodpecker's habitat. A solution was found by financing protection of woodpeckers on private land and providing regulatory assurances. Safe Harbor Agreements spread rapidly across the bird's habitat and as of 2020 there are more than 400 landowners conserving 2.5 million acres of woodpecker habitat.

In 2006, the population had grown to 6,105 active clusters and the bird is expected to be delisted by 2030.

¹²³ Leopold, 1934, 537-544

¹²⁴ Kraus et al, 2021, 1088-1127

¹²⁵ Holmer, 2016

¹²⁶ Baier L, 2020 'Saving Species on Private Lands, PERC, <https://www.perc.org/2020/05/06/saving-species-on-private-lands/>

Spotlight: Candidate Conservation Agreement with Assurances

CCAA apply to species that are not yet listed. Under such agreements a landowner can agree to carry out activities to conserve a species with the assurance that, if the species is ever listed as Endangered or Threatened, the landowner will not be subject to any additional regulations imposed.

CCAA are advantageous in that they provide landowners with immediate funding for species conservation, instead of having to wait for the species to be listed, and long-term assurances about land use restrictions.

According to the US Fish and Wildlife Service, early conservation of candidate species generally increases management options and flexibility for landowners, lowers the cost of recovery for the species, and reduces the likelihood that more restrictive land use measures will be necessary in the future.

Canada and Australia also provide for conservation agreements between the Government and any person to provide for the conservation of a species or habitat. Aboriginal organisations in Canada have entered into such agreements with the federal Government.

Incentives for habitat and species protection, are common internationally. Canada and Australia both have a stewardship programme. Under SARA, the Minister can establish a stewardship action plan that creates incentives and other measures to support voluntary stewardship actions. A stewardship action plan must include commitments to:

- Regularly examine incentives and programmes that support actions taken to protect at-risk species;
- Provide information and increase public awareness about at-risk species;
- Sharing information about at-risk species, including community and aboriginal traditional knowledge;
- Create awards and recognition programmes;
- Provide information respecting programmes related to stewardship agreements, land conservation easements and other such agreements; and
- Provide information relating to the technical and scientific support available to persons engaged in stewardship activities.

In 2015, a report on Canada's Habitat Stewardship Programme for Species at Risk revealed that it was not fulfilling its mandate because of a lack of implementation of SARA, specifically, a lack of critical habitat identification and incomplete recovery instruments were hindering strategic stewardship investment.

In Australia, habitat stewardship payments are offered to private landowners through a tender process. For example, the programme paid incentives to landowners to implement five-year management plans to improve the quality of red-tailed black-cockatoo habitat in a defined catchment area.¹²⁷

European countries have a multitude of stewardship schemes for protecting species and habitat, including agri-environmental programmes required by EU regulations. As an example, Ireland introduced a rural environmental protection scheme in 1994 which sought to, *inter alia*, provide for

¹²⁷ Garnett et al, 2018, 149

the protection of wildlife habitats and endangered species and fauna. The regime is currently under review and is expected to change from an area-based scheme to a results-based scheme. The scheme is being partially funded by a carbon tax.¹²⁸

Other financial incentives are also used overseas to encourage conservation by private landowners. For example, conservation easements to reduce tax burdens and tax shifting, whereby a tax increase on non-priority land compensates for a tax relief on high-priority land, which is provided in exchange for conservation activities on that land.

The U.S.' conservation banking programme allows landowners to protect habitat for endangered species then sell conservation credits to developers to mitigate habitat alteration elsewhere. For example, the Hickory Pass Ranch Conservation Bank in Texas placed a conservation easement on several hundred acres of a 3,000 acre ranch to protect Golden-cheeked Warblers, in exchange for credits that can be sold to mitigate impacts to Warbler habitat elsewhere.¹²⁹

Lessons learnt: Incentives

1. The protection and recovery of threatened species requires assistance from private landowners.
2. Internationally, private land conservation is encouraged using a range of financial incentives.
3. Incentives are an important part of preventing perverse outcomes and getting private landowners to engage with threatened species laws and protections.

Key drivers of transformational change

This comparative review of threatened species laws has been undertaken with a view to identify the tools and mechanisms necessary to ensure a transformational change in the protection and recovery of Aotearoa's most imperiled species.

Often a tool or mechanism looked like it could result in transformative change for a species or its habitat. Invariably though, further investigation into its effectiveness was disappointing. Primarily, this was due to poor implementation resulting from the influence of vested interests or economic considerations.

The following is a list of key drivers required to move towards transformation change. These drivers should be read in conjunction with the lessons learnt detailed in the green boxes above.

Give primacy to species

A key driver of transformational change is giving species primacy, not only under wildlife law, but also more broadly in land use decision-making. The lack of commitment to prioritising species in decision-making is a key implementation challenge brought about by vested interests. In many cases overseas it hindered the effective application of processes and mechanisms that had the potential to make a transformative change for species and their habitat. Incentivising good behaviour, penalising the bad and thwarting opportunities for vested interests in threatened species decision-making is key to ensuring species primacy.

Prioritise the most threatened species

¹²⁸ <https://www.agridirect.ie/article/rebs-2-must-deliver-for-farmers>

¹²⁹ Holmer, 2016

Regulation does not need to be applied equally to all threatened species. The degree of control can be staggered down the threat continuum. Prioritising the most threatened species is not to say that unlisted species should be left unprotected. Pre-listing conservation measures can be a far more cost-effective and positive way to manage biodiversity.

Give independent bodies decision-making powers

Ministerial discretion and political interference in decision-making is a significant impediment to the effective and efficient implementation of threatened species laws. Giving decision-making to an independent body, insulated from political interference, including in the appointment of its members, would remove that impediment. Decisions made by an independent scientific body could include:

- What species are prioritised for assessment
- The extinction risk ascribed to a species
- Identification of critical habitat and residences
- What recovery actions are required
- Whether to grant wildlife permits

Independent bodies overseas currently make recommendations on the above matters. For example, COSEWIC is a widely respected independent body that recommends the threat status of a species and Hawaii's Endangered Species Recovery Committee is a partially independent body that makes recommendations on permits and other agreements. However, these bodies do not make final decisions. In a knowledge-based system, they should.

Ensuring that decisions are informed by the best available science and indigenous knowledge is a natural corollary of independent decision-making. In Canada, dual reporting of scientific knowledge and indigenous knowledge is informing extinction risk assessments.

Provide for greater automation

A corollary to independent knowledge led decision-making is that things should happen automatically. For example, as soon as a species is assessed as threatened, it is listed as such. Automatic listing has been heralded as a way of improving threatened species laws overseas. Automatic listing expedites the process and means that all species which qualify for listing would, in fact, be listed. Adding to the robustness of species lists by eliminating political interference in listing decisions.

The requirement to automatically designate a species' critical habitat when that species is listed is a significant strength of threatened species law in the U.S., and automatic habitat protection upon listing is a main reason why Ontario's system was so highly regarded. However, both jurisdictions have struggled with effective implementation. Political interference and economic considerations have hindered identification in the U.S. In Ontario, an overzealous exemptions regime has undermined protections.

Adopt a precautionary approach

Decision-making in the context of uncertainty was one of the five main interventions identified by the IPBES to generate transformation change in biodiversity loss. Scientific evidence relating to threatened species is usually incomplete and uncertain. In the face of such challenges, a precautionary approach should be applied to threatened species decision-making. A precautionary

approach is particularly important in the context of climate change which is exacerbating and multiplying threats to threatened species in new and dynamic ways.

A precautionary approach might mean that for example:

- Data Deficient species are afforded some degree of protection until further information comes to light proving they are not threatened.
- Critical habitat is protected immediately upon listing, to be refined later.
- Adverse effects on the most severely threatened species and their habitats are avoided.
- Adverse effects on other threatened species and their habitats that cannot be avoided, minimised or remedied, are prohibited, rather than allowing for offsetting or environmental compensation.
- Climate change is considered in all decisions affecting threatened species, even when the effects remain unclear.

Protect habitat

Land use change and associated lack of habitat protection is driving biodiversity decline. Thus protecting threatened species habitat is vital. Protection of critical habitat and/or residence can be required via threatened species legislation. A robust incentives scheme will greatly assist in reducing opposition to habitat protection.

Require an overall net gain

A key tenant of giving species primacy is ensuring that threatened species decision-making provides for an overall net gain to the listed species. This means that any permit or plan approved or developed will leave the listed species better off than if that permit or plan did not exist. It requires a mindset of going 'above and beyond' for the species, rather than just ensuring no net loss.

The rationale for an overall net gain requirement is that biodiversity decline is significantly advanced, globally and in Aotearoa. Significant changes are required to reverse that trend. It is no longer good enough to simply maintain species and their habitats. To bring transformative change, we need to be proactively enhancing and restoring populations and habitats. The need is particularly urgent in depleted environments.

Impose greater accountability

The lack of accountability imposed on decision-makers is a common criticism of threatened species law. Accountability goes hand-in-hand with transparency and, as discussed above, broad Ministerial discretion is causing issues. Threatened species decision-making should be free from political influence and knowledge based. Decision-makers that digress from that knowledge need to be held accountable for outcomes that follow.

Commentators in Australia and Canada have called for an independent auditor or review body responsible for reviewing the performance of decision-makers. For example, in Australia, an independent statutory office has been recommended to regularly review the performance of federal, state and territory governments in meeting Australia's international and national responsibilities for biodiversity conservation, including the identification and abatement of major threats.¹³⁰ It is suggested that the position could be a parliamentary commissioner or an inspector-general (as for federal biosecurity in Australia).

¹³⁰ Invasive Species Council et al, 2022

An ability for civil society to hold decision-makers to account through judicial review proceedings is still likely to be an essential part of the accountability picture. In Canada, a specialised tribunal for at-risk species disputes has been recommended by academics as a way of improving the Canadian system and keeping the Government accountable under SARA.¹³¹ SARA contains numerous statutory timeframes, compliance of which is critical to avoid Canadian species decline. An external tribunal could ensure that the Government complies with the law, and that delay does not creep into the system. New Zealand's Environment Court was referenced as a good example of a similar institution working in practice.

The heart of threatened species law is to prevent species from becoming extinct. But what happens when that occurs? Currently, the answer is nothing in the jurisdictions we reviewed. Species can fade away without any legislative repercussion. This seems unacceptable for Acts designed to keep species alive. The difficulty in enforcing accountability for extinctions is the challenge of proving a species is in fact extinct. Directing accountability to an increase in threat status might be more appropriate.

Increase funding

Underfunding is a major barrier to the protection and recovery of threatened species. No matter how well crafted a threatened species law is, it cannot achieve its objectives unless it is sufficiently financed.¹³² This funding needs to be comprehensive, targeted and directed upon certain actions (i.e., listing of a species).

Australian legislation provides no assurance of adequate funding with which to implement recovery plans or undertake conservation actions.¹³³ For example, upon listing, the EPBC Act does not automatically allocate funds to the protection or recovery of that species. The country's main environmental grant programme, 'Caring for Our Country', has rejected recovery funding for listed species which has been a significant contributor to their eventual extinction.¹³⁴

The U.S. has a much better record of species recovery than Australia, primarily because funding for actions is mandated under the ESA. Based on the U.S.' average spend per listed species (AU\$1.2million), Australia would need to spend AU\$1.8 billion a year to achieve recovery of its 1,800 species. It currently spends on average just AU\$70,000 per nationally listed species.

A recent estimate concluded that effective threat abatement and species recovery in Australia would likely require targeted funding of at least AU\$1.5 - 2 billion a year.¹³⁵ Linking threatened species plans to funding arrangements is key to ensuring their success. For example, Australia's TSSC has recommended that the EPBC Act be amended so that upon listing of a threat, a costed plan be agreed under a National Environmental Threat Response Agreement.

Hawaii has a statutory 'endangered species trust fund' which collects money from numerous sources, including:

- Money accrued from the sale of retail items officially sponsored by the department for the fund;

¹³¹ Turcotte et al, 2021, 1474–1494

¹³² Westwood et al, 2019, 136–160

¹³³ Woinarski et al, 2016, 13-23

¹³⁴ Woinarski et al, 2016, 13-23

¹³⁵ Invasive Species Council et al, 2022

- Private contributions for the management and recovery of Hawaii's unique plants and animals;
- Fees and assessments charged for the use of public land and waters and designated for the fund;
- Penalties, fines, or auctions resulting from enforcement violations;
- Legislative appropriations; and
- Money deposited to implement the obligations of a habitat conservation plan, as security for habitat conservation plan funding, or technical assistance program fees.

Ensuring the recovery continues

A significant issue with many recovery programmes is ensuring that the species or habitat is protected long-term. The successful recovery of many listed species is dependent on ongoing and often intensive recovery actions, such as pest control. Once this work discontinues, the population is at risk of reverting to its previously vulnerable state, undoing all the hard work invested in achieving its recovery. Secured long-term funding is required. Ideally, this would be targeted at a landscape-scale, rather than a species level.

The glossy black-cockatoos demonstrate this tenuous recovery position. Since the mid-1990s glossy black-cockatoos have undergone a significant revival on Kangaroo Island in Australia primarily because of intensive possum control. It is highly likely that its population will again decline if possum control funding ceases. Ecosystem-scale reduction of possum numbers would reduce the need for such intensive management but that is likely to remain unresolved for the foreseeable future, not least because the possum is a native to Kangaroo Island.

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